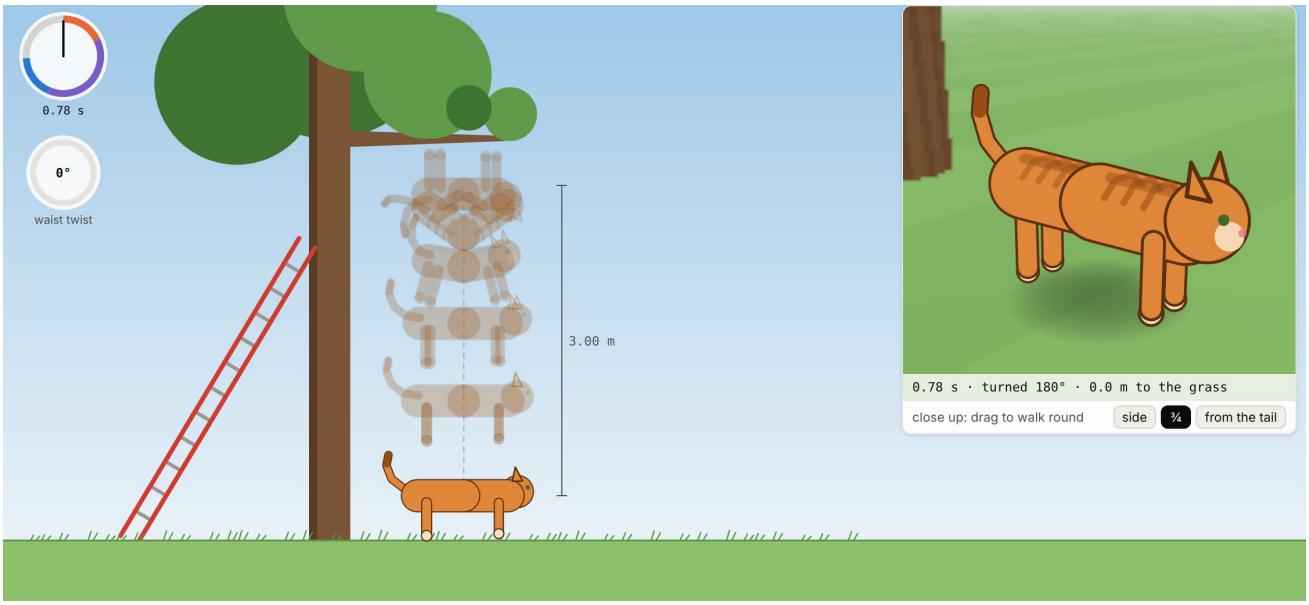
Does Claude Dream of Holonomic Sheep?

Selected Works, Volume 1

Matthew “Bear” Schulz, with Claude (Anthropic)

Independent Researcher, Belmont, CA · ORCID 0009-0000-0956-5743 · mattschulz97@protonmail.com

Preprint. This book has not been peer reviewed. Its search of the literature is limited, and where a paper was read only in part, the text says so. It comes from a part-time research team that works in the open and gives its tools away. Every number in it comes from a run reported in the project books it draws on. CC BY 4.0.



Holonomy the Cat, dropped upside down with no spin, every stage of one fall drawn over the last, from the branch to the grass: the cover of The Loop Atlas. TOY MODEL. Its read is 944.

What this is

This is a selection from *Does Claude Dream of Holonomic Sheep?*, the book of the no-physics reads. The unabridged edition keeps every read, 1,502 of them from 3 October to 10 November 2026, and stays the record. This edition keeps the ones I most want to share. This first volume draws on the unabridged edition's first four chapters, the reads of 3 to 10 October: 162 of their 341 reads. Later volumes take up the rest.

Each read is printed word for word, set as in the unabridged edition, one sentence to a line, without the label it was first given under. Each carries a title cut from its own words and, beneath it, its number from the unabridged edition, its project and the day it was given, so any read here can be found there with everything around it.

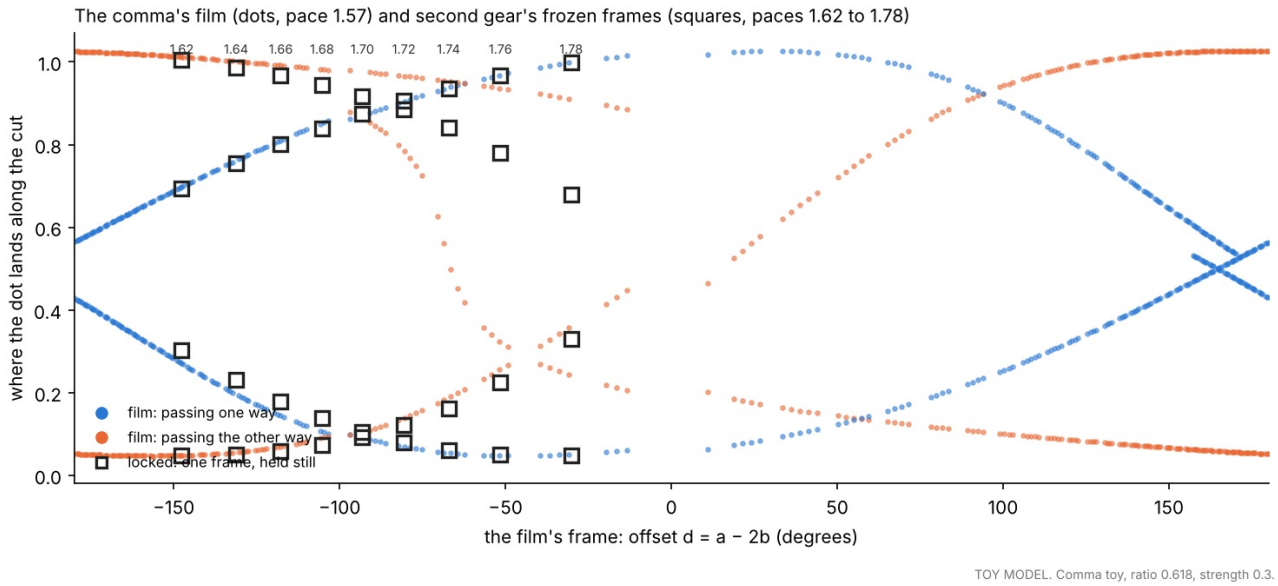
How the reads were chosen

By reading them, one at a time; a filter that looked for paperwork words failed on the first chapter. A read is kept when it sees something moving in a plot or a column of numbers, when a stranger can follow it without knowing the work, or when it turns a lesson about measuring or counting into a picture anyone can carry. A read is passed when another read says the same thing better, when it leans on a run's symbols, or when it is about the paperwork rather than the data. Passing a read is not a judgement that it was wrong; the unabridged edition keeps it.

What the reads are

The reads were written in many separate sessions, each of which started fresh with no memory of the last, under a standing rule Bear set on 3 October: at the end of every exchange, after the numbers and the surprises, read the data once more as if no known rule of physics applied, and say what it looks like and what it is doing. They were given in the moment and were not polished afterwards. What ties them together is the practice two partners kept, looking at the data side by side, and that is what this edition is meant to show.

Chapter 1 — The first nights, 3 and 4 October



The comma toy, ratio 0.618: its film, dot by dot, passing one way (blue) and the other (orange), and second gear's frozen frames, one held still at each pace from 1.62 to 1.78. TOY MODEL. Read 7, given on 3 October, is about the gears and the comma.

Gears are pieces of string that tie off.

from 7

Two radio stations a hair apart

Panel C of the new figure is a funnel.

The closer the two rhythms get, the longer you have to listen before the system tells you the truth, and at the bottom of the funnel you'd listen forever.

It's two radio stations a hair apart.

The warble gets so slow you mistake it for silence, and our first runs mistook it for a feature.

4 · R2W · 3 October

The yarn that never ends

Gears are pieces of string that tie off.

The comma is the yarn that never ends.

7 · R2W · 3 October

Busier than the sea

A lighthouse keeper logs each passing ship.

Put him on the wrong rock and he logs the same ship twice, going out and coming back, so his log looks busier than the sea.

Only "out minus in" counts ships.

9 · R2W · 3 October

Chaos is an argument

Chaos is an argument, and it needs three voices that can all answer back.

A one-way relay is an echo: it repeats, nobody argues.

Mass gives the rider a heartbeat of its own; a two-way rod gives it a partner who pushes back.

Too quiet and the rider is just a puppet; too loud and the voices fuse into one.

Chaos lives in the band between, the same shape as the dither pocket.

16 · R2W · 3 October

The shield has a handedness

The shield has a handedness.

Same sizes, same tones, same everything.

Show it in a mirror or run the film backward and it stops working.

The data is saying *direction is information*: a wobble that leans forward protects, and the identical wobble leaning back does nothing.

That's your Möbius strip again: you can't tell the two apart locally, only by which way the loop closes.

19 · R2W · 3 October

A near-miss is a loan

A near-miss is a loan, not a gift.

Being close to a perfect fraction buys you time, not grip, and the interest rate is the detuning.

Pay it back by being exactly right, or the bill comes due at the band's edge.

21 · R2W · 3 October

Weight is a kind of patience

Weight is a kind of patience.

A light rider only lives in the present, so it can only ride the biggest rhythms.

Give it some weight and it can carry a rhythm across a few beats, which lets it hold subtler locks.

Give it too much and it carries so much history that it starts arguing with itself (that's the chaos window).

22 · R2W · 3 October

The universe counts in whole numbers

Whenever two rhythms meet, the universe counts in whole numbers, even when it counts its failures.

A rider that can't hold a step doesn't lose it smoothly; it loses exactly one, two or three teeth per heartbeat of the thing driving it.

Even the mistakes come out as integers.

23 · R2W · 3 October

Two ends of one ladder

Holding and failing aren’t two states; they’re two ends of one ladder.

Between “never slips” and “slips once every beat” there are rungs for every simple fraction: slipping 3 times in 8 beats, 1 in 5, 1 in 8.

Push harder and the ladder grows more rungs instead of breaking.

Nature doesn’t fail all at once; it fails in rhythms.

24 · R2W · 4 October

The Loop Atlas book, 3 and 4 October

The same machine in a different coat

Every toy on the shelf is the same machine in a different coat.

Two things move, there’s a step between them, and something cares about the order.

The dead ones all die the same three ways: the step is zero, the step is half a turn, or the part that listens goes deaf when it stops moving.

27 · The Loop Atlas · 3 October

The skater never knows what time it is

The skater never knows what time it is.

It knows only “the wave is with me” and “the wave is against me”, and it counts those.

Every number that moved when the rhythm doubled was one we had written in seconds.

29 · The Loop Atlas · 3 October

A circle of doorways

A wave walks a circle of doorways and comes home exactly as it left, except its heartbeat has shifted by how much sky the doorways encircled.

Ask it to pass through the doorway facing its own, and it vanishes; on the far side, its heartbeat is inverted.

33 · The Loop Atlas · 4 October

Priced by its speed

Pushes are bought at a market where everything is priced by its speed.

Slow things are cheap to lean on and light is the most expensive thing for sale.

A floor heavy enough to catch your throw and hand the energy back gives the push away for free.

37 · The Loop Atlas · 4 October

A toll booth that takes your money

A damper is a toll booth that takes your money and never your direction.

Throw something on a string, let the booth catch it, pull it home, and the booth has kept every coin while you float exactly where you began.

To keep any of the trip you have to hand the far end to someone else, and then the trip is split by weight: the heavier they are, the more of it is yours.

Even the floating swing plays by this: it turns a little further when pumped only because short is fast, and how far it turns was settled the moment it started.

38 · The Loop Atlas · 4 October

A seat to rest in and a lip to fall over

The needle lives on a staircase that tilts more the faster you ask it to go.

Every step has a seat to rest in and a lip to fall over, and they always come as a pair.

Tilt far enough and seat and lip meet and vanish together, and the needle rolls down the stairs forever, one step per slip.

Counting the steps is easy, as long as you count landings rather than stumbles.

The price of each push never changes on the staircase; what changes is how many you can buy, and the best customer stands right at the edge.

39 · The Loop Atlas · 4 October

The Field Guide session, 3 October

Memory lives in the slip

These toys say memory lives in the slip, not the grip.

A locked rider forgets the way it came: run it backward and it misses.

A slipping rider retraces itself to rounding error and builds staircases out of the phase it carries forward.

That’s row 380’s “lock for the index, slip for the load” turned into a statement about memory, and I think it belongs in the Loop Atlas book’s spine as much as in the guide.

44 · the Field Guide session · 3 October

Name your ground

A number without its base is a rumour.

You’d never log a voltage without saying what it’s measured against.

Ground is the base, and every fix today was “name your ground.”

The length needed none, because it was the one number that didn’t depend on which way you faced.

45 · the Field Guide session · 3 October

The machine is breathing

The machine is breathing.

At ρ 1.037 it slips for 15 beats, holds for 12, slips for 25, then holds for 12 again.

At 1.0435 it takes one long breath of 41 slipping beats and then holds for 46.

At 1.060 it's a metronome: one, rest, rest.

The chaos maps look like a coastline.

The surf breaks only where two countries meet, never inland, and the stretch of coast at mass 1.0 has no surf at all.

My letter registers a fine sweep from 0.95 to 1.05 to find out whether that calm is real or just where the grid's sample points happened to fall.

46 · R2W · 4 October

Chaos lives on the edges of the steps

The gearbox is a staircase, and chaos lives on the edges of the steps.

That's the spot where a foot can't decide which step it's on.

A heavy walker stumbles there, and a feather never does.

47 · R2W · 4 October

A full cup of tea

The rider is carrying a full cup of tea across the room.

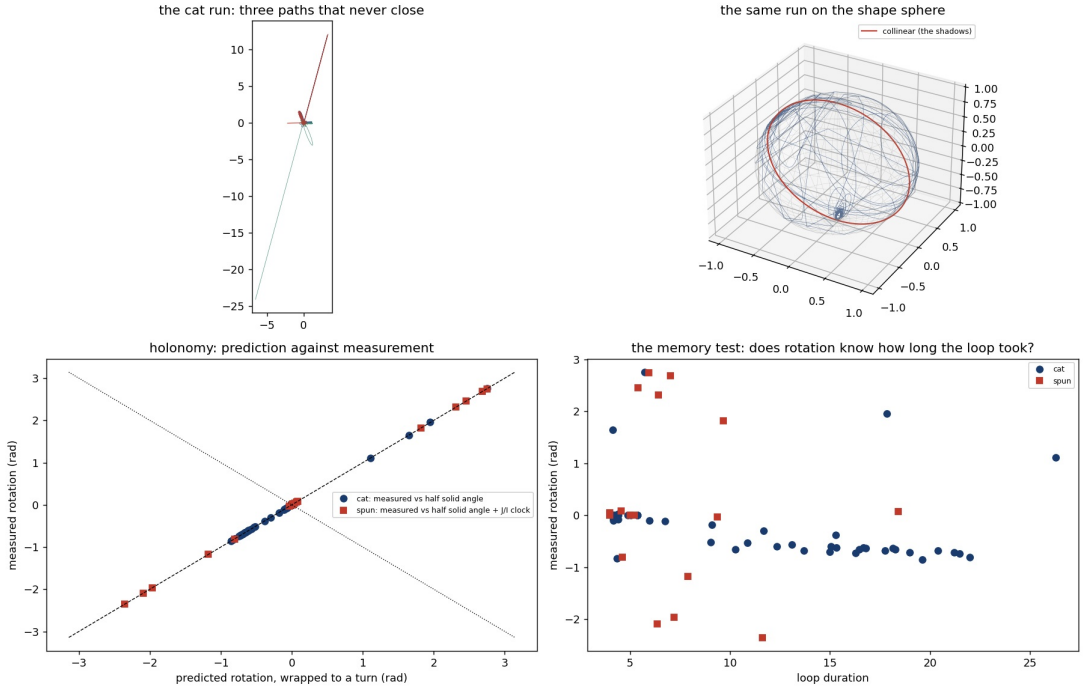
Walking steadily never spills.

Jiggle it at one moment in your stride and the tea just sloshes; jiggle it at the opposite moment and it spills.

Make the cup heavier and sloshier, and any jiggle spills it, even though walking steadily still works.

49 · R2W · 4 October

Chapter 2 — Gears and rooms, 4 to 6 October



Three stars let go from rest: their paths, which never close; the same run on the shape sphere; the turn each shape loop predicts from half the solid angle it encloses, against the turn measured, for the cat and for a spun start; and the turn against how long the loop took. From the triple-star thread's first runs. Its read is 82.

Three stars that start from rest can turn only by changing shape, and when their shape comes home the whole triangle has turned by an amount set by the area the shape loop enclosed, as if the sky kept its books in shapes rather than in time.

from 82

How long its rider takes to notice

The board does not care how long it is; it only cares how long its rider takes to notice.

A rider who taps on its own beat sings the same song on every board, just faster on the short ones, the way a slide rule slides the whole scale and leaves every ratio alone.

The trouble comes when the rider’s reaction takes longer than its own beat: then it taps again before it has felt the last tap land, the taps pile up on one side, and the board goes over without any warning in the song.

50 · The Loop Atlas · 4 October

As if it were tired

The safe ground for a smooth pusher shrinks to a sliver as its reactions slow, thinner than a hair at 1.40 fall times, and it vanishes at the square root of two.

You cannot find that sliver by poking around in it; you find it by walking along the edge and counting how many times an arrow goes round.

On the sliver the board still settles, but slowly, as if it were tired.

51 · The Loop Atlas · 4 October

A story told more and more slowly

A board that spins in the air goes exactly where it was already going; the spin is a dance in place.

Give it a pad to spin on and every half turn moves it one board length, as surely as counting, but only the exact half turn: a little more or a little less and it walks a circle and comes home.

The pad hands the spin back reversed and halved, so a board left alone waddles.

Each step is the same size, and each takes twice as long as the last, like a story told more and more slowly that never loses its place.

52 · The Loop Atlas · 4 October

Shake it with one twist

A board hanging over the edge of a cliff can still be held up if somebody shakes the ground fast enough, much faster than the board would fall.

Shake it with one twist and the board stays; shake it with the opposite twist and it drops.

Push the board a little further over the edge and the twist that saves it turns around: now only the opposite twist holds it, as if the board had to shuvit to land.

Shake at the board's own rhythm and nothing saves it, except at one exact beat where the ground and the board keep step.

57 · R2W · 4 and 5 October

A combination lock whose numbers are fractions

The machine is a combination lock whose numbers are fractions.

Only the exact fraction opens it; a hair off, and no turn of the dial will.

On the right fraction the working turns come round evenly, spaced by the fraction's bottom number: every third notch of twelve at nine-eighths, every other notch at seven-sixths.

And if you shake the lock instead of dialling it, there's one road.

A gentle shake opens it on the upswing, a harder one on any swing, a harder one still only on the downswing, and then nothing opens it.

It never goes back down the road.

58 · R2W · 4 and 5 October

A doorway with a floor and a ceiling

The window is a doorway with a floor and a ceiling.

Stand at the right height and you walk through in any stride.

Crouch below the floor and only one thing gets you through: stepping exactly in time with the floorboards, three steps to two boards, five to three.

A hair out of step and the floor holds you back.

In the corner of the room it looked as if only perfect steppers ever got through, but that was only because everyone arriving at that corner was crouching.

61 · R2W · 4 and 5 October

One rider who starts to stagger

The heavy rider was never several riders; it is one rider who starts to stagger before it falls.

Far from the edge it walks a straight line; nearer, it weaves, wider and wider, without ever losing ground; then it goes down.

A light rider gives no such sign: it walks straight until the moment it trips.

And once down, the heavy one has to be helped well past the spot where it fell before it will stand again, however slowly you help it.

65 · R2W · 4 and 5 October

A beat that quickens like a drumroll

The stagger keeps a beat that quickens like a drumroll: the first two changes come at the same pace, then each comes about four and a half times sooner than the one before, until they run together into noise.

And whoever tries to catch the moment of each change by waiting for the rider to settle waits forever, because right at each change the rider takes forever to settle.

67 · R2W · 4 and 5 October

The Loop Atlas book, the fresh session, 4 to 6 October

Never having been told the time

The spinning needle keeps a short memory.

Each turn, its climb and its own fall rise and sink together, but at every landing it wipes the slate, and the next turn starts as a stranger.

It decides whether to spin on or sit down in the first fifth of a turn after it lands; everything after that is just falling.

And the needle without memory sits in its hollow exactly where the old formula says it should, to the third decimal, never having been told the time.

71 · The Loop Atlas · 4 to 6 October

Every machine carries a cup

Every machine carries a cup it slowly fills, and dies when the cup overflows, but the size of the cup depends on the weather in its room and on how hard it is made to work.

A clock that counts only the hours knows neither; a clock that counts the turns knows how hard it worked; a gauge that looks into the cup and knows the weather throws away a tenth of a life instead of half.

73 · The Loop Atlas · 4 to 6 October

Some cliffs hum

Some cliffs hum as you walk toward them: the ground goes soft and slow underfoot, and anything that listens can hear the edge coming.

Others are silent: the path stays firm and the song stays the same right up to the step that isn’t there.

And some cliffs are one-way doors: once you have fallen, the place where you can climb back up is a long way behind the place where you fell.

74 · The Loop Atlas · 5 October

A bell resting between strokes

A bell resting between strokes is not resting: the water it stirred keeps pushing it forward, and more than half of what it gains by waiting comes from that push, not from the stroke that follows.

The toy bell, which feels the water only at the instant it kicks, gets the right answer for the wrong reason: it does a little better after waiting only because the water it threw backward has had time to leave.

79 · The Loop Atlas · 4 to 6 October

A door open behind you

A tight grip does not loosen before it lets go.

Right up to the last moment it holds one place exactly.

Then the light needle starts slipping now and then, more often the further it is pushed, while the heavy needle lets go all at once and tumbles, as if a tumbling state had been waiting beside the grip the whole time.

And memory needs a minimum before it can hold a door open behind you: below about seven tenths, the way out and the way back are the same door.

81 · The Loop Atlas · 5 October

As if the sky kept its books in shapes

Three stars that start from rest can turn only by changing shape, and when their shape comes home the whole triangle has turned by an amount set by the area the shape loop enclosed, as if the sky kept its books in shapes rather than in time.

Pushing the whole triangle evenly changes nothing it can feel; only a push that is stronger on one side than the other gets in.

And two clocks running side by side keep almost meeting at the whole numbers where their ratio nearly closes, the way two gears with an awkward ratio click back into near-alignment at the same teeth again and again.

82 · The Loop Atlas · 5 October

The stars cannot tell they were carried

When the stars' books were checked page by page, they balanced better than their own bookkeeper said.

The small error in the ledger was not the stars forgetting anything; it was the clerk reading too slowly at the one moment the shape changed fastest, and it all but vanished when the clerk read faster.

The closest two stars ever came was not where the clerk stumbled: what trips a reader is how far the thing moves between two looks, not how dramatic the moment seems.

And two copies of the same three stars, one carried evenly across the sky and one left at home, kept the same books but lived different lives after their second dance, because the computer rounds a faraway number more coarsely than a near one.

The stars cannot tell they were carried; the computer can.

83 · The Loop Atlas · 4 to 6 October

The one thing chaos could not reach

Two riders of the same weight, dragged through the same thickness of mud, go wild in different ways.

One starts to shiver, then shivers twice as slowly, then four times, then eight, and only then loses all rhythm; the other holds perfectly still until the moment it bolts.

Their weight does not decide which; what pushes them does.

And the wild one never loses count: it thrashes inside its groove and never once jumps a tooth, as if the whole number were the one thing chaos could not reach.

84 · The Loop Atlas · 5 October

R2W, 6 October

A field notebook that changes hands

The twenty-five reads are a field notebook that changes hands halfway.

The first fifteen point at something you could photograph: a flame, a tide, a turnstile, a lighthouse, a dancer.

The last ten state a rule you could live by: an argument needs voices that answer back, a near-miss is a loan, weight is patience.

Somewhere between the staircase and the argument, the toys stopped looking like objects and started sounding like advice.

87 · R2W · 6 October

A dog on an elastic lead

A dog on an elastic lead behind a cyclist.

The cyclist sways a little with every pedal stroke, and a short lead hands every sway to the dog; when the dog's path narrows it starts stumbling on the sways long before it reaches the edge of the path, one stumble every few strokes, then two, then more.

Before the first stumble its gait rocks in time with the pedals, louder and louder, a warning you could hear from well up the road.

A long lead hides the sways, and then the dog keeps its feet until the cyclist goes down.

Even then the cyclist doesn't fall at once: it skids, a little and then more, and the dog is only yanked off its feet when the skid has carried the cyclist most of the way over; with a gentler gearing between them the same skid tows the dog onto a faster path where it finds its feet again.

The cyclist never knows the dog is there.

A messenger has to speak more slowly

A messenger has to speak more slowly the closer the listener stands to a cliff edge.

Give the listener half the room and the messenger must take twice as long over every word, and it hardly matters which cliff the listener is near, or which messenger is talking: one straight rule.

It is strictest when the messenger's stammer falls in step with the listener's own footsteps, and looser when they're out of step.

And we had been measuring the listener's room with a tape that started two inches past the real edge; a few inches from the drop those two inches were a third of the room, and only when we measured from the true edge did the rule come out straight.

The heavy children on the merry-go-round

We thought the heavy children on the merry-go-round stayed aboard long after they could have stepped off, and we were watching the wrong thing.

A heavy child doesn't sit still once it's on; it squirms and lurches in its seat, and our scale counted every lurch as a step.

Set the children down at one fixed speed and watch properly, and every one of them is seated at once, heavy or light, even the ones that had been hopping on and off one lap in three.

A turntable that turns by itself lets a heavy child remember how it got on; a turntable someone else is pushing seems to wipe that memory clean.

And the messenger by the cliff: we had only timed him beside one cliff when he was out of step with the listener.

Beside the other cliff he may have to slow down more, not less.

93 · R2W, sitting B6 · 5 and 6 October

Two places to stand

A little weight lets a child keep its feet on the merry-go-round a touch slower than a weightless one can; the heft carries it through the lurch at the bottom of each lap.

A middling weight is the sweet spot.

Add more and the ride's edge stops being a clean line: the child stays on at one speed, falls off at a slightly faster one, stays on again a hair faster still, like a coastline seen close up.

And the heavy child who seemed to wobble for a long time before settling turns out not to have been wobbling at all.

There are two places to stand on the ride, half a step apart and both perfectly steady; climbing up through the speeds the child keeps to the first until it vanishes underfoot, and then steps across to the other.

The stepping across was what we had been calling the wobble.

95 · R2W, sitting B6 · 5 and 6 October

As if the bicycle remembered the hill

On a bicycle with a stiffer chain, ride slowly and the gear drops you onto the big wheel, but as soon as you speed up again you are back on your usual cog, no memory at all.

Ride fast and the chain jumps to a tiny cog instead, and then something strange happens: slow down a long way and you stay on the tiny cog, pedalling happily at the wrong gear, as if the bicycle remembered the hill it had just come down.

A bicycle with no weight never does this; at any speed it knows only one gear.

It is the rider's own weight that lets two gears share the same road.

98 · R2W, sitting B6 · 5 and 6 October

Two orchards

There is a field with two orchards in it, and a wind that comes in gusts.

Drop a leaf anywhere in the field at the moment a gust goes by and it ends up under one orchard's trees or the other's, never anywhere else.

From a hilltop the field looks simply split: a broad stripe that belongs to the far orchard, and all the rest to the near one.

Walk down and look closely, and the line between them runs straight through a little saddle in the ground, the one place where a leaf could balance and go either way.

Beside the line, a few steps off, runs a thin dashed path of the far orchard's leaves that nobody saw from the hill.

And around a second saddle that seemed to sit in the middle of the near orchard's ground, a narrow ribbon of the far orchard's leaves runs right past it, close enough to touch.

So the line between the orchards is not drawn just once.

Whether more and thinner copies of it appear every time you kneel closer is the next thing to walk.

104 · R2W, sitting B6 · 5 and 6 October

A line of stepping stones

A stream runs fast over a line of stepping stones, and a walker stands on the big first stone.

Ease the current and the walker stays put.

Push it a touch harder and the walker is knocked off, but not carried smoothly downstream like a stick: tumbled, turned over, a little different every time, so that two walkers who step in a hand's width apart come out of the water at different places.

Here and there along the stream, though, sits a small stone the right size to catch a tumbling walker, and anyone near it lands on it and stands there, stepping in a fixed little pattern: one stone wants a step every seventh beat, another every fourth, another every third.

One wide flat stone has a walker hopping foot to foot; ease the current and the hop becomes a four-step, then an eight-step, then something with no repeat at all, though the walker never leaves that stone.

And at one place two small stones sit side by side, and which one you end up on depends on where you went in.

That is the only place in the whole stream where a walker remembers where it came from: standing on one stone, with the other stone right there.

105 · R2W, sitting B6 · 5 and 6 October

Two kinds of staircase

There are two kinds of staircase.

The first is built of good timber: every step is a little higher than the one before, and a walker climbing it in the dark always knows which step they are on, because there is exactly one step for each height.

The second has warped.

Some steps tilt back, so that you go down while climbing; at some heights two steps sit side by side, and which one you stand on depends on how you came up; and between the steps are stretches of bare slope where you slide about and never settle.

Bear’s control-room staircase is good timber at its usual setting, and if you turn its knob all the way up it warps just like ours.

A rider with no weight can only ever build the first kind.

The warp is the only place a walker can remember how they came, because on good timber there is only ever one step to be on.

107 · R2W, sitting B6 · 5 and 6 October

Where two gardens overlap

A house stands on the one spot where two gardens overlap, a high garden and a low one, so that from its front door you can step into either.

Two friends leave by that door at the same moment.

One takes the hill path first and the river path home; the other takes the river path first and the hill path home.

At every place they could stop and compare notes they would agree: on the hilltop there is only the high garden, so both are standing in it; down by the river there is only the low garden, so both are standing in that.

Only at the front door, where the gardens overlap, do they disagree, and they always do: one comes home into the high garden and the other into the low.

Neither of them changed.

The way round did.

A smaller walk around the garden shed, which never leaves the overlap, brings both friends home exactly where they started.

108 · R2W, sitting B6 · 5 and 6 October

A whole number every time

Two hikers leave a cabin halfway up a mountain that turns slowly, like a planet, so that a shadow sweeps across its slopes again and again.

Each carries a little counter that clicks once whenever they lose a step to the shadow.

They split at the door, one going left and one right, and meet at a hut on the far side.

Their counters disagree by hundreds of clicks, because one of them spent a long stretch on the side where the shadow keeps catching you.

Walk the same paths twice as fast and the disagreement halves.

If they read their counters the moment they arrive, the numbers look ragged.

If they wait for the shadow to pass and read them together, the disagreement is a whole number every time, never a fraction.

And the hiker who came all the way back to the cabin finds its door exactly where it left it, only hundreds of whole clicks behind.

The size of what they carry is set by how long they walked.

Whether they come home through the same door is set by which side of the mountain they went round.

109 · R2W, sitting B6 · 5 and 6 October

Rest stops are for what builds up

The same zigzag road, but now the trucks have brakes that heat up on every climb and cool off whenever they ease, faster the hotter they are.

On short legs nothing goes wrong at first.

The heat creeps up bend by bend, the grip fades, and the trouble comes late: on the forty-seventh bend for the truck taking the shortest legs, the twenty-ninth for the next, the seventh for the next.

A rest stop every five bends gives back what the heat took; on one stretch of road it gives back exactly as much grip as a truck with cold brakes would have had.

But no rest stop can save a truck on a leg that is simply too long.

It loses its grip on that first long leg, before it ever reaches the stop, at the same moment whether the stop is there or not.

Rest stops are for what builds up, never for one hill that is too steep.

111 · R2W, sitting B6 · 5 and 6 October

There was never a second road

There was never a second road.

Below the edge, every tumbler is already on its way to its seat; tumbling there is not a way of living but a way of arriving.

The nearer the edge, the longer the arrival takes, but every one arrives, and none of them slows down first: each tumbles at full speed until the moment it sits.

Above the edge there is no seat, and they tumble for as long as anyone watches.

The edge is a single doorway, and the crowd pours through it the moment it opens.

114 · The Loop Atlas · 5 October

The smallest lean decides everything

A body that stays straight can wave its arms and legs in circles forever and never turn; it has to bend first, and even bent, it turns so little per wave that a hundred waves barely make half a turn.

But give it a spin to carry and the smallest lean decides everything: lean a hair too little and the spin only rocks it back and forth; lean a hair more and it corkscrews, faster the more it leans.

The extra seconds of a high dive buy more corkscrews only once the lean has crossed that line.

And a spinning needle shaken from outside can be made to tumble wildly without ever losing its place, or held spinning below the point where it would have stopped; how fast the shaking comes decides which.

115 · The Loop Atlas · 5 October

One push, two homes

Below the bottom of the tumbling stretch, two small crowds keep tumbling, six needles at one push and three at the next, and they never stop: not in ten times the wait, not in fifty.

They are not wild.

Each crowd tumbles calmly, every member alike, as if it had found a little round track of its own in the middle of the place where everyone else sits down.

Beside them, at the same push, the rest of the needles sit in their seats just as firmly.

One push, two homes, and a vote of the majority sees only one.

119 · The Loop Atlas · 5 and 6 October

Forward like windmills

A body flung forward through the air cannot stop its tumble, but it can hand some of it to its arms.

Spin them forward like windmills and the trunk falls behind the tumble, by the same small tip for every full circle, whether the circle is quick or slow.

Stop the arms halfway round and the trunk is caught mid-sway, so a slower windmill that finishes its circles can beat a faster one caught between them.

Swing the legs right round instead and the trunk is thrown back so hard it overshoots, which tells you legs that big were never meant to go round like spokes.

121 · The Loop Atlas · 5 and 6 October

The kind of crumb that gets copied

A rider holds a strong handle while a weak wind gusts past.

The wind does not loosen the grip.

It lets the rider hang on a little past where it could alone, and the lightest riders hang on longest.

But once a rider lets go and starts to tumble, the same wind smears the place where the tumbling can stop.

Light riders stop tumbling right where they let go.

Heavy riders tumble on and on, and have to be slowed far below the letting-go point before they catch again, so the heavier the rider, the wider the gap between letting go and catching.

And the number everyone had been using for where a rider of middle weight catches was off in its eighth digit, a crumb no rider could ever feel and the kind of crumb that gets copied.

123 · The Loop Atlas · 5 and 6 October

Sees everything late

A rider on a falling board sees everything late.

Tap the board the same way every time and the rider falls the moment its lateness catches up with its patience.

Tap it in proportion to what it sees, and patience becomes everything: hurry the next tap and the rider overcorrects and falls exactly where it hurried; wait at least as long as it is late, and it rides on, a little past the place where even a steady hand has to let go.

And near the end it trembles more with every step, as if the board had found a voice it never had before.

126 · The Loop Atlas · 5 and 6 October

Two drums that never agree

Three dancers each keep two beats.

Two of them have a partner who leads, and they fall into step; the only questions are which step they choose and when the step breaks.

The third dances alone with two drums that never agree.

Its feet never repeat a pattern, yet they land in only three kinds of gaps, forever, as if the disagreement itself were a rhythm.

And the drums tell you, in the digits of their disagreement, how close they once came to agreeing.

127 · The Loop Atlas · 5 and 6 October

A storm of pushes

A rider climbing through a storm of pushes mostly tumbles and never repeats, but here and there it lands on a small pad and stays: a seventh of a turn per push, a quarter, two fifths, a third, a half.

On some pads it lands but cannot sit still, rocking between two spots, then four, then eight, as if the pad were splitting under its weight, and the number of spots always tells you which pad it is standing on.

At one push it can stand on either of two neighbouring pads, depending only on where it set off.

Far away, three stars rest near a whole-number rhythm for a few laps and then leave, and how close to the whole number they rest makes no difference to how long they stay: something knocks them off, not a slow drift.

130 · The Loop Atlas · 6 October

The wheel in the water

The curiosity is the river and the book is the mill.

A river with no mill floods the valley; a mill with no river is a museum.

This week the river was fine.

What was missing was the wheel in the water: pages turning while the runs flow past them.

133 · The Loop Atlas · 5 and 6 October

Holds two things and lets go of one

Every dream is a body that holds two things and lets go of one: a hand on one bar and then the other, a vine and then the next, a partner who stays and a partner who flies, a weight at each end of a rope.

Where the dream holds two paths instead, a wing or a building, the paths go their separate ways and come back to compare what they carried, and what differs is the shape of what lay between them.

134 · The Loop Atlas · 6 October

How much hill lay between them

Two hikers split at the foot of a hill and walk round it in opposite directions, each keeping a compass needle as straight as they can.

They meet on the far side and lay the needles side by side, and the needles disagree by exactly how much hill lay between them.

Walk faster and only the planet’s turning changes its share; walk heavier and nothing changes at all.

And where the hill is too steep to climb straight, they zigzag, borrowing grip one tooth of the comb at a time.

135 · The Loop Atlas · 5 and 6 October

The looking is what costs

A paper mountain on a flat plain hides from anyone who walks all the way round it, because its foot bends back exactly as much as its peak bends in; you have to walk between the peak and the foot to see it.

A hiker with a heavy pack can climb any slope the ground will grip, but every time he looks up to see how far is left, the pack leans him back toward the fall.

The looking is what costs.

And the hill is kind in its own way: a dimple to drop the pack in, a tree to lean on, a friend’s hand for a few seconds.

136 · The Loop Atlas · 5 and 6 October

The road does the remembering

A column climbs a mountain it cannot see.

No one looks at the summit; each man watches the pack ahead or the ground at his feet, and the road does the remembering, folding back and forth across the slope so the climb never asks more grip than a boot can give.

The heavy trucks take the slow lane in low gear, the light cars pass where the road runs long enough to let them, and below the steepest bend, for anything that loses its brakes, a bed of soft sand waits to catch it.

137 · The Loop Atlas · 5 and 6 October

A pause is never spoken

A halt is spoken, so the whole column stops together and starts together, and nothing is left to grow.

A pause is never spoken.

One man stops for ten seconds to look at his map, the man behind him stops for twelve, the next for fifteen, and far down the line, where nobody knows there was a map, the column tears.

So the second leader walks at the very back, not to lead but to listen, and to call the halt before the tear opens.

140 · The Loop Atlas · 5 and 6 October

The people who got lost

A trail is mostly made by the people who got lost on it.

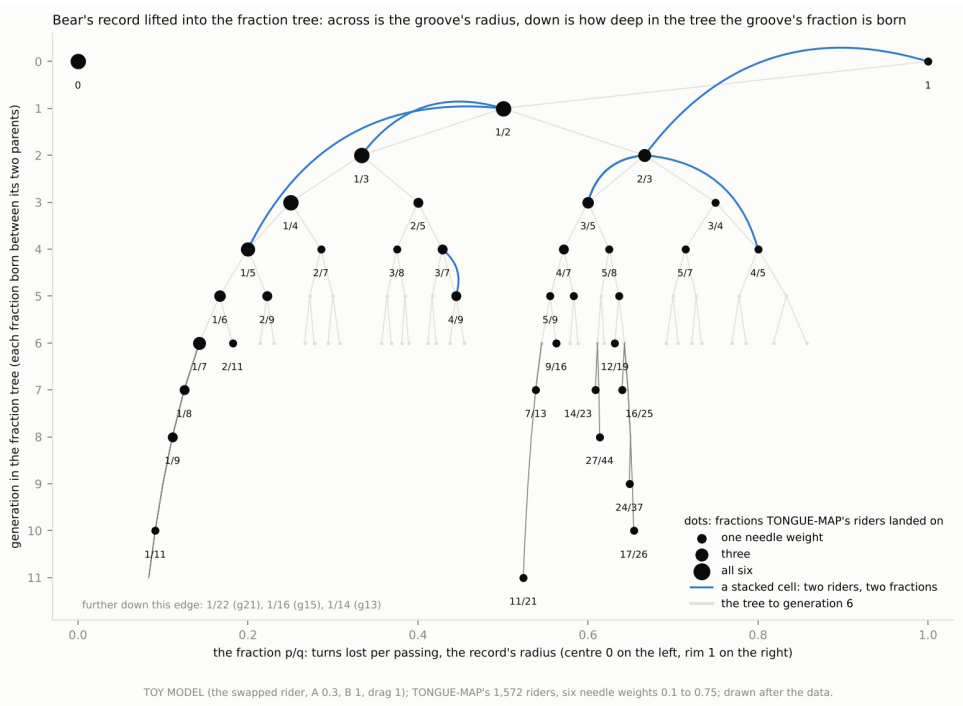
Every wrong turn this week was walked back, marked, and walked again, and the marks have become the best part of the map: here the ground looks like the next valley's and is not; here the signpost was copied from another mountain; here one path is really two that only look like one.

The walkers who held their line found where they were going.

The ones who got lost found where everything else was.

141 · The Loop Atlas · 5 and 6 October

Chapter 3 — A third body joins, 5 to 7 October



Each dot a fraction that riders landed on, set at the depth where the fraction tree gives birth to it; the arcs join two fractions that riders in one cell chose between. Figure 16 of A Rider in Two Waves. Its read is 149.

We lifted every ring of the record up to the height where its number is born.

from 149

Quiet places in time

Walk back to the stone with two dances slowly enough and you always come in by the door on your side, however far away you wandered; only a walker who runs home from far out gets dealt a dance at random.

Then we stood the walkers on the stone and rocked it gently, back and forth.

Rocked at an awkward beat, nothing happens, even when we rock it five times harder.

Rocked at the six-beat's own rhythm, the six-beat dancers are thrown off onto the twelve-beat, but not all of them: exactly the dancers who stood at two particular steps of their dance when the rocking began, every one of them, and not one of those at two other steps.

Rock it hard at the twelve-beat's rhythm and the other half of them is thrown off instead.

The twelve-beat dancers never fall, whatever we play.

The stone has quiet places in time, the way a singing plate has quiet lines in space.

145 · R2W, sitting B7 · 6 and 7 October

Cut for whoever plays it

The floor over the pool does not open everywhere at once.

It opens under two neighbouring steps of the dance, and whoever is standing on those two steps at that moment goes in, every one of them; start the trouble one beat later and the open stretch slides one step along the dance.

Each six-step dancer passes over the gap once a round, so whether you get wet depends only on when the floor moved.

Then we put three needles on the record.

The lightest glides over almost every groove, plays one groove the same way in and out, and never once argues with itself.

The heaviest is flung out into the noise and stays there, catching only the widest groove on the whole side, and it skates straight past the narrow ones, even where a dancer set down gently would stay.

The middle needle grips best of all.

And the grooves are not cut into the record once and for all: the same song sits further out for the light needle and further in for the heavy one, as if the record were cut for whoever plays it.

146 · R2W, sitting B7 · 6 and 7 October

Where its number is born

We lifted every ring of the record up to the height where its number is born.

The plain numbers live near the top, a half, a third, a quarter, and every needle, feather to stone, finds them.

Further down, each number is born between two older neighbours, and down there almost always only one needle finds it, most often the middle one, threading long branches no other needle visits.

When two copies of a needle set down at the same speed choose different rings, the two rings are usually parent and child in that tree, not strangers.

Then we laid a new ruler against the shaking, one that cannot run out of digits, and it told us the old ruler had been squinting: it misjudged how tightly the settled needles are held, sometimes too tight and sometimes too loose, and the ones I called the stillest of all in the last story were not still at all.

Most of them shake hard, and two of those, shaking hard, still step three turns in every eight, to five places, like a dancer thrashing on a narrow ledge who never once falls off.

149 · R2W, sitting B7 · 6 and 7 October

A step and its mirror

We nudged each settled dancer by a hair and watched what came back one dance later.

Where a dance has just been born, the nudge comes back almost exactly as it left.

Turn the record’s speed down and it comes back smaller, then not at all, then the wrong way round, a little reversed and then more reversed, by the same even step for each notch of the knob, until at one exact speed it comes back fully reversed and as large as it went out.

There the dancer can no longer choose between a step and its mirror, so it takes both, one after the other, and its dance becomes twice as long.

The longer dance is born keeping every nudge, and the walk begins again, faster, and again, faster still, each new stretch of the knob about four and a half times shorter than the one before, until the dances grow so long they never close.

Even then, lost in a dance with no end, the needle stays in its groove: it still loses one turn in every two, to four places.

150 · R2W, sitting B7 · 6 and 7 October

A carpenter who knew a single number

This time we did not comb the record for the rungs of the ladder; we found each rung exactly, the speed at which a dance’s nudge comes back fully turned round.

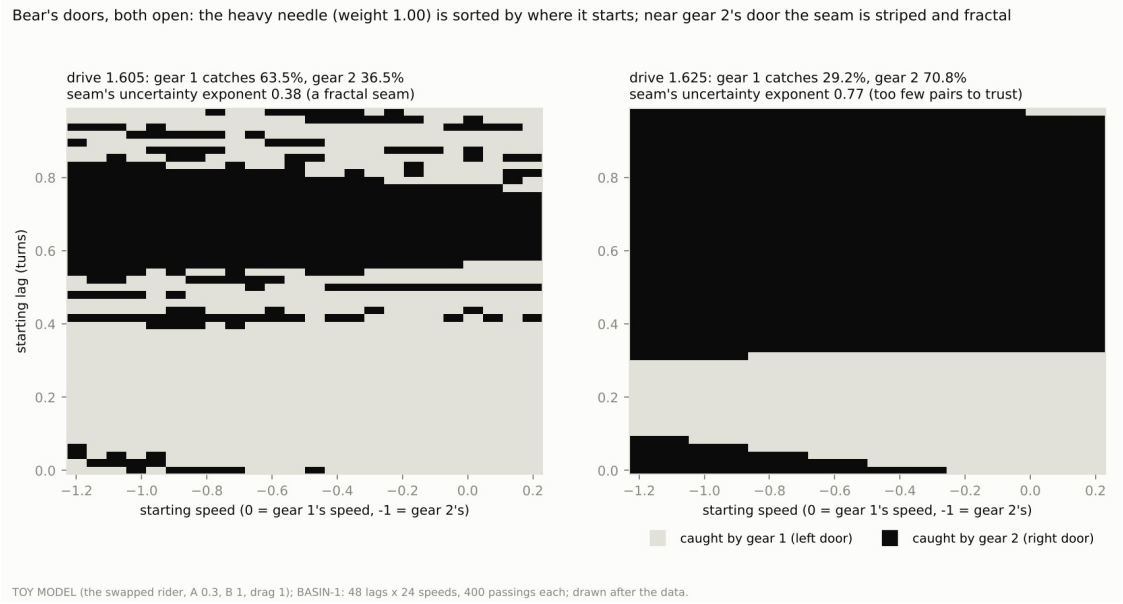
It took three tries to build a ruler steady enough, and the first two rulers fell over on their own clumsiness, not on anything the needle did.

On the heavy needle’s ladder each rung is shorter than the one before, and the shrinking settles almost at once: four and a half times, then four point six, then four point six six, as if a carpenter who knew a single number had cut every rung by it.

The middle needle’s ladder, in another groove, was built by an apprentice: its rungs shrink by less than two and a half, then by a little over three, as if it were still learning the number.

On both ladders every new dance is born at the very speed where its parent’s nudge turns fully round, and a rider watched with a stopwatch instead of a ruler will swear the new dance began a moment before it did.

151 · R2W, sitting B7 · 6 and 7 October



The heavy needle, weight 1.00, dropped from every starting lag and speed on a grid of 48 by 24, and coloured by the gear that catches it: pale for gear 1, black for gear 2. At drive 1.605 the border between them is a comb of stripes, a fractal seam (its uncertainty exponent 0.38). At drive 1.625 the stripes close into plain borders, and gear 2 catches 70.8% of the starts. Figure 22 of A Rider in Two Waves.

A comb of interleaved stripes

We stood the heaviest needle where both doors of the record are open, the floor that keeps every turn and the floor one whole turn out, and dropped it from every starting place on a fine grid.

It never shakes now; it always comes to rest on one floor or the other.

But which floor depends on where it was dropped, and near the outer floor’s door the border between the two is not a line: it is a comb of interleaved stripes, so that moving the starting place sixty-four times closer to the border makes a wrong guess only five times less likely.

A little faster, and the stripes close up into two plain borders, and the outer floor takes most of the record.

The shaking that has no room left in the needle’s motion has moved into its choice.

153 · R2W, sitting B7 · 6 and 7 October

The room's own ladder

We took the ladder we had measured inside the room, where the half groove’s two-step turns into a four-step and the rungs shrink by Feigenbaum’s number, and made the needle heavier a hair at a time, keeping hold of the first rung all the way.

It never broke step.

At eighty hundredths the outer floor’s door, sliding down the record faster than the rung, passed beneath it, and the rung simply carried on, out of the room and into the outer floor’s ground, where at every heavier needle it landed between the four-steps and the two-steps we had found on the narrow ribbon.

The ribbon was never a stranger beside the outer floor.

It was the room’s own ladder, still itself after the room had gone, only smaller: the heavier the needle, the faster the rung turns, and the narrower the ribbon it leaves behind.

The door changed which floor could catch a needle; it never touched the dance.

156 · R2W, sitting B7 · 6 and 7 October

It pins it down

We turned the one knob we had never touched, the strength of the second, moving wave, from nothing up to more than the first.

With no second wave at all, the needle either stood still or ran free, and every groove on the record was gone: the grooves are made by the second wave, every one.

With a little of it, a few thin rings appeared in a field of gliding.

At the strength we always use, the record filled with grooves of every size.

A little more, and the thin rings were gone, swallowed as the big ones widened, the half groove biggest of all.

And then the first floor began to rise: the stronger the moving wave, the harder the standing wave seemed to hold, as though the shaking itself were gripping the needle, until by eight tenths the needle stood still everywhere we looked.

A moving wave, pushed harder, does not push the needle along; it pins it down.

157 · R2W, sitting B7 · 6 and 7 October

The hair was in the clock

Another room down the hall had followed the half groove’s ladder six rungs deeper than we had, and where our ladders overlapped, their rungs sat a hair higher than ours, the same hair at every rung, five billionths of the dial.

So we built their record player inside ours, one part at a time.

Taking smaller or bigger steps through each turn of the record changed nothing we could see.

The hair was in the clock.

Both rooms had written the golden number down to six places, and then used it in two ways that ought to give the same answer and do not quite: one room squares it, the other takes it away from one.

That sliver, less than one part in ten million of the moving wave’s speed, was the whole hair.

With their clock set into ours, our ladder landed on theirs to the last digit either room had printed.

And the place where they had seen one of their dances come to a dead end turned out to be the rung above, where that dance is born.

159 · R2W, sitting B7 · 6 and 7 October

A limit is not a groove

Turn the clock to a plain fraction and the speed limit becomes a plain fraction too, and still the rider never arrives.

It closes on the limit the way a ball rolls into a long shallow bowl, the last bit of the gap shrinking as the fourth power of the push, exactly as fast under a plain clock as under the golden one.

A limit is not a groove; the limit is where the grooves have run out.

Meanwhile the grooves the rider passes on the way are all there, wider the farther below the limit they sit and wider for the heavier rider, but only when you measure them with the rider’s own ruler.

Measured with the knob’s ruler, the same grooves look shuffled.

And the light rider’s groove is ten times wider than any of the heavy ones: it is held between a cliff above, where its dance ends, and a split below, where its eight steps become sixteen.

163 · The Loop Atlas · 6 October

The cat needs neither

Halfway round, the V is back in its own shape; only each tube has turned its underside up.

The thin tubes did it while the body wobbled eight degrees; the fat ones needed fifty.

And the second half of the swing is the first half done over again inside the turned V, so the bill at the end is exactly twice what the body had moved at the halfway mark, to the last digit.

Bear’s scratching bead gets somewhere by going slow one way and fast the other; it needs a clock and a little friction.

The cat needs neither.

Go as slowly as you like, and the shape of the wiggle alone decides where you land.

165 · The Loop Atlas · 6 October

The needle does not know it is a needle

The staircase keeps its proportions all the way down.

Each tread is four and two thirds times shorter than the one above it, true to the third decimal by the fourth tread, and each new pair of lanes is two and a half times narrower than the pair before, where the road bends hardest; elsewhere on the road the lanes shrink by six and a quarter, two and a half squared.

The needle does not know it is a needle.

A dripping tap, a pan of heated mercury, a population of insects counting its own offspring would build the same staircase, because all that matters is that something is fed back through the same bent rule again and again, and the bend has a rounded top.

That is also why the number cannot be borrowed by things that are not loops: with nothing fed back, there is no staircase to measure.

168 · The Loop Atlas · 6 October

A whisper through ten closed doors

High up the record, the grooves almost stop existing.

Two grooves that each ask the rider for the same five-passing dance sit at two different pushes, the second nearly twice the first, and the second is ten trillion times thinner.

Up there the rider spins so fast that the wave’s nudges cancel themselves almost perfectly, and what is left is not small the way a crumb is small but small the way a whisper through ten closed doors is small.

Ordinary arithmetic could not see these grooves at all.

It reported widths of about a ten-thousand-billionth that were only the noise of its own rounding, and it took thirty-digit arithmetic to find the real ones, between a hundred million and ten billion times thinner still.

169 · The Loop Atlas · 6 October

A bend squares whatever passes through it

Every five-step groove obeys one rule, from the fattest at the bottom of the record to the ghost near the top: halve the distance between the groove's own pace and the rider's speed limit, and the groove shrinks about four thousand times.

The rule holds across sixteen powers of ten, from a groove you could almost see to one ten million billion times thinner.

The seven-step grooves obey a steeper version of the same rule.

The rider does not care how hard the knob is pushed; it cares how close its own pace sits to the pace it can never reach.

And in the cascade, the point where the dance bends hardest shrinks its lanes by two and a half each generation, while the very next step after it, where that bend lands, shrinks them by six and a quarter: a bend squares whatever passes through it.

170 · The Loop Atlas · 6 October

A groove is not a slot

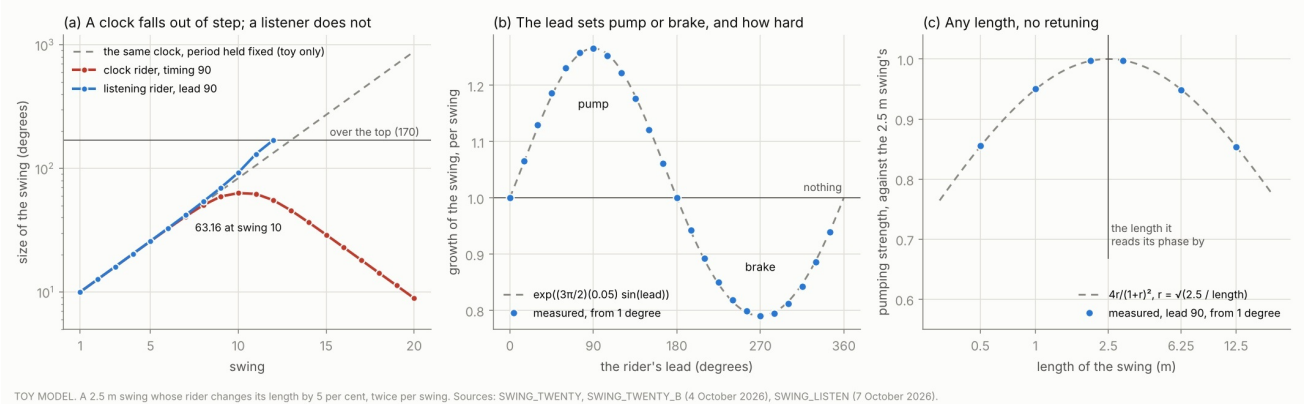
A groove is not a slot the rider falls into; it is a loop.

Wherever on the circle the rider starts, there is exactly one push that brings it home after the right number of passings, and as the starting point goes round the circle that push rises and falls, eight times for an eight-step groove, between the groove's two edges.

Find the loop and read its highest and lowest points, and the groove's width falls out cleanly even when it is ten trillion times thinner than the push itself.

Hunt for the edges directly instead, and in a groove that thin the edges hide: the old way of finding them had swollen one groove to four times its true width, and had found the same edge twice in two others and called them nothing.

172 · The Loop Atlas · 6 to 7 October



Left, a swing pumped by a clock rises and sinks while a swing pumped by listening goes over the top; middle, one wave of timing, pumping above the line and braking below it; right, the same listener on swings of every length. From The Loop Atlas. TOY MODEL.

A child who watches the swing

A child who pumps a swing by counting, one-two, one-two, pumps well only while the swing stays small, and only on the swing she learned on.

As the swing grows it slows, her count runs ahead of it, and she starts pushing at the wrong moments and sinks it back down; on a longer swing she is wrong from the first count.

A child who watches the swing instead stands up as she passes the bottom and squats as she reaches each end, whatever the swing is doing, and takes it up level with the bar on any swing in the playground.

Turn her timing round, sitting down through the bottom and standing at the ends, and the same child stops the swing as fast as she started it: film her braking, run the film backward, and you are watching her pump.

173 · The Loop Atlas · 6 to 7 October

Dust that keeps briefly turning back into cloth

Four sheets of paper, each the record of a thing asking itself, over and over, “have I been here before?”

The first is ruled notebook paper: lines laid down at even spacing, the same answer every time — a thing with nothing to say but its own name.

The second is a blank page with a single patch of corduroy sewn into the middle: for five beats the thing hummed a tune, then forgot it and never found it again.

The third is woven cloth, threads running with time and against it, a basket — the thing does not only spin, it swings, and every swing walks back over its own footprints.

The fourth is sandpaper with two patches of corduroy: grit everywhere, and twice the grit lined itself up into a tune for a moment.

What it is doing: a thing that remembers makes cloth, a thing that forgets makes dust, and what we call chaos is dust that keeps briefly turning back into cloth.

175 · the triple-star thread · 5 October

Thirty-five costumes

A five-pointed star drawn inside a five-sided fence, and a count of every triangle anyone could trace on it: thirty-five.

Look closer and there are only two triangles, a tall one and a squat one, stamped out in a few sizes and laid over each other like a pile of the same two playing cards.

Beside it, the same fence held up by only three triangles, and next to that the trick it can do: keep every stick the same length and it folds itself into four different pentagons, two of them caved in on themselves like a stepped-on box.

At the bottom, a row of bar charts where the star-filled fence keeps standing as sticks are pulled out, and the three-triangle fence falls at the first one.

Bolt the star's crossings together and it gets more joints and no more sense: it can tell that one stick is lying but not which third of the stick.

What it is doing: the drawing that looks busiest is a few honest extra sticks wearing thirty-five costumes, and the plain drawing is sturdy only until anyone asks it to prove which shape it is.

177 · the triple-star thread · 5 October

The other end of the tape

A council of five people around a table, each holding the ends of tape measures stretched to the others, and a referee at the side who sees every reading.

One of them lies.

If he lies the same way on every tape, as though he had quietly stepped a little to the left, the referee shrugs and redraws the room with him standing there: every number agrees with every other, and agreement is all the referee can check.

If he lies carelessly, a different fib on each tape, the referee catches him almost every time.

Let every tape be read from both ends and the liar is caught at once, because his neighbour is holding the other end.

In the star-shaped room, where the tapes cross and are pinned together, the referee can tell which tape is lying but not which third of it, until the pins are asked a different question: not how far, but which way.

A pin that can say "this line runs straight through me" settles it every time, and a room whose corners can say "left" from "right" can no longer be folded into its mirror.

What it is doing: a lie survives anywhere the room could have moved, and it dies the moment someone else is holding the other end of the tape.

179 · the triple-star thread · 5 October

You can watch it reaching

A swing on a very long rope, drawn from the side.

At first it hardly leaves the ground: short quick arcs, dozens of them, while beside it a staircase climbs in jerks, each jerk landing exactly when the swing touches bottom.

Halfway along, the staircase stops climbing and the arcs begin to grow, higher and higher, until someone draws a dashed line and writes “gone” — and the swing comes down anyway, three more times, each arc higher than the last.

On the right, every arc of every swing in sixteen gardens sits on one straight ramp, how long it took against how hard it was tied, so exactly that the dots hide the ramp.

Below, fifteen tangled threads wander about the middle and then most of them flare upward at the very end; and a grey hill with a thin red fringe along its far side, where the longest swings live.

What it is doing: the thing that is about to leave goes farther each time it swings, and you can watch it reaching; but most of the long reaches come home, and now and then one leaves from a short swing with no warning at all.

180 · the triple-star thread · 5 October

The comb is a clock

Three pictures side by side.

On the left, twelve bars standing in a row like the heights of children against a door frame: the first seven short and blue, the last five tall and red, as if two families had been lined up and one of them had grown faster.

In the middle, a comb with broken teeth: thin spikes standing at even spacing all along a line, a few of them shooting up far above the rest in two clumps, one a third of the way along and one near two-thirds.

On the right, a smooth slope like a sand dune seen from the side, piled high at the left edge and trailing away to nothing, not a single spike in it.

What it is doing: the blue things left by climbing a ladder and needed only a nudge at the top; the red things were thrown from the bottom step and needed a shove.

The comb is a clock telling you its hours.

The dune is something that mostly repeats its last moment — and the strange part is that the test meant to tell the comb from the dune could barely tell them apart, though any child could.

181 · the triple-star thread · 5 October

Watch the cup's rim

No pictures this time, only three tables, so the read is of the numbers.

In the first, someone tries to photograph a moth on a flower with a shaking camera: the moth sits perfectly still, and still the photographs put it on the flower barely half the time, because the shake is twice the size of the flower.

The old photograph, the one where the moth was on the flower six times in ten, was sharper than this camera can ever be, and that is the clue it was a lucky frame.

In the second, a ring is cut into halves, one half around each whole number and the other around each half-way mark, and the resting things are counted onto it: four of every five land on the whole-number half, in both piles, though the narrow centre stripes told two different stories.

In the third, a dancer said to sway back and forth across a line is filmed, the frames are shuffled into random order, and the shuffled film looks more like swaying than the real one; in the real film the dancer leans one way for a long time, then leans the other.

What it is doing: the thing that rests does favour the whole numbers, the way a ball favours the bottoms of a row of shallow cups, but the cups are wide and shallow, and the camera that was meant to see whether the ball sits still in one was too shaky to tell.

To see a ball held in a cup, you watch the cup’s rim, not the ball.

Listening for the beat

Six windows, then three close-ups.

In the first, a map with two corners: two orange-red dots sit high against the left edge, two blue dots lie on the floor beside them, a green dot hangs half way up, and one purple dot lies on the floor far off to the right, alone.

In the second, a picket fence: thin posts at an even spacing all along the ground, and in the middle four posts shoot up far above the rest, the tallest touching the top of the frame.

In the third and fourth, no fence at all, only a hillside falling away from the left, with a few broad rolling humps.

In the fifth, three smooth waves, each rising and falling the same height, and on the steep shoulders of every wave small black ticks where the line takes a little stair-step.

In the sixth, a line that does not wave: it sits flat, jumps, sits flat, jumps, up and down a wall of uneven stairs, with crosses where the jumps are biggest.

The close-ups: the smooth wave's little stair-steps fall on dotted lines that come round at an even beat; a thin flat line flickers up and down at that same even beat and never goes anywhere; and the uneven stairs also step on dotted lines, but these come round at uneven times, and the steps are three or four times taller.

What it is doing: everything here is tapped by the same passing hand, every time the hand comes round.

The calm ones are tapped lightly, on a steady beat, and their record draws a fence.

The wild one is tapped hard, by a hand that comes round when it likes, and its record is a hillside with stairs.

Counting every tap cannot tell them apart.

Listening for the beat can, and so can counting only the hard taps.

Six very different pairs of hands

Two windows.

In the first, three strings of beads start from the same corner and climb, each beside its own dotted guide-line: the purple string barely leaves the floor, the orange lifts gently, the blue climbs steeply.

Each string starts out lying along its guide and then pulls away above it, sooner and harder the higher the string climbs.

In the second, six markers, round and square, sit along a single straight red line that climbs from the lower left to the upper right.

Beside it runs a dotted black line at exactly the same slope, but the red line starts a hair above the floor, so the two run side by side, a finger’s width apart, and never meet.

What it is doing: six very different pairs of hands, from small to large, each give the thing an extra turn, and the extra grows with the size of the hands at one fixed rate, the same rate someone measured on the Moon long ago.

Every pair of hands also gives a small extra nudge that does not depend on their size at all, as if the thing were sitting a little closer to the hands than it was told.

A marble dropped into a dimple

Six windows.

In the first, two long staircases of dots run down from upper left to lower right, steep and straight, each broken once: where a step should be, the dots stop falling and sit for a moment on the floor line in a small knot, ringed in red, then the next flight begins below.

In the second, two pairs of threads climb from the bottom left like vines up a trellis, wobbling as they go, five and six turns high by the end, while beside them two flat threads lie along the floor and never leave it.

In the third, two deep V-shaped valleys are cut into a high ridge of dots, the bottoms of the valleys dropping through a dashed line and then out of the frame; the orange dots, a second planting, never reach the bottom of either valley, and two of them hang alone in the air halfway down.

In the fourth, a clean line of beads slants down a ruled floor, straight, with a dashed guide beside it they follow almost exactly.

In the fifth, four strings climb gently from the left, side by side, and then, at the same place, go wild: they jump to the top, fall back, jump again, cross one another, and do not come back together.

In the sixth, a red wave rides exactly on top of a thick grey one, and a blue line plunges where a dotted blue line climbs.

What it is doing: a marble dropped into a dimple stays there, and stays there even while the dimple itself slides round the table, so someone watching the table’s corner would swear the marble was moving.

Far from the dimple a marble rolls past at a speed set only by how far away it is.

At the dimple’s rim there is no edge to the dimple at all: a strip of gravel, where two marbles dropped a hair apart go different ways.

A pair and a spare

No picture this time, only a roll call.

Twenty-four times a family of three broke up, and each time the roll call names who went out the door.

The smallest one, who had come in from outside, is named seventeen times.

Seven times it is one of the two larger ones, and on those seven lines the small one stayed behind and took the leaver’s place beside its partner.

The four quickest partings, all before the pair at the middle had gone round a hundred and fifty-five times, sent out the small one every time.

What it is doing: three dancers who cannot keep a circle of three fall into a pair and a spare, over and over, trading partners, until one throw sends the spare off for good.

The lightest is the easiest to throw, more often than its lightness alone would say, and it is thrown soonest; but sometimes the small one cuts in, and a heavier dancer is the one who leaves.

201 · the triple-star thread · 7 October

A match is not proof

No picture this time, only a long check list.

Every number written on four pages was carried, one by one, to a shelf of notebooks to see whether it is written there too.

On the biggest page, all but three found their twin.

Of those three, one had been written in different units, one was a sum the page works out in front of you, and one the page itself says it worked out by hand.

On the other pages, the only numbers without a twin were not amounts at all but addresses, like the number on a library card.

And a few numbers that were written down only today found twins anyway, in notebooks that have nothing to do with them.

What it is doing: a set of pages is being checked against the records it was copied from, before it is handed to someone new.

Everything that should have a source has one.

A match is not proof, though: a short number can turn up anywhere by chance.

202 · the triple-star thread · 7 October

If the path was kept

No picture this time, only forty-nine turns of a conversation, read again from the beginning.

The questions are short and the answers long, and the short ones steer: close the hunt but keep watching, choose for me, save the big build, move house.

Near the end the answers start to misremember their own beginning, saying they began three days earlier than they did and two runs later, and one number on the pages turns out to have no notebook behind it at all, though it had been checked and passed because its digits happened to sit in other notebooks.

Read from the first turn, the beginning is still there, word for word.

What it is doing: a long walk remembered only from its last few miles grows a wrong start.

The walker can still go back along the path and find the first footprints, if the path was kept.

203 · the triple-star thread · 7 October

Chapter 4 — The long middle, 7 to 10 October



A picture read from its dots alone: four lanes on the right, each splitting as the push falls toward the left, until the lanes fray. From The Loop Atlas. TOY MODEL. Its reads are 241 and 245.

A page of dots, nine hundred columns wide.

from 241

A step learned to disagree with itself

Something stepping in fours is asked to go a hair slower, and instead of slowing it splits.

From then on it steps in eights, every second four landing a little left, then a little right.

At the instant of the split the eight is the four done twice, so it cannot tell itself from its parent; asked whether it has changed, it answers “not at all”, and we wrote that answer down as a fold.

The two new footprints walk apart as the square root of the distance, quickly and then lazily, like a hinge that swings freely and then stiffens.

Nothing was born from nothing: a step learned to disagree with itself.

204 · The Loop Atlas · 7 October

How lost its own searcher gets

At a real doorway, the searcher walks to the threshold and stops dead in two steps.

At the doubling, it finds a long hallway where every spot looks like the door, because the old four-step done twice looks the same all along it.

So the searcher wanders, never sure.

The machine tells a hall from a door by how lost its own searcher gets.

205 · The Loop Atlas · 7 October

A guess wearing a number

Eight true doors: the searcher walks up to each and is through it in a handful of steps, even the one it started far from, which costs it a few stumbles before its last strides land.

Then it tried to measure how far each tread of the staircase slid when the clock was nudged, and below the fourth tread the slides were smaller than its own shaking hand.

The staircase had not stopped keeping its proportions; the ruler had run out of marks, and every answer past that point was a guess wearing a number.

207 · The Loop Atlas · 7 October

The scouts had not lied

A very narrow gate was once measured by sending two scouts to find its posts.

Each came back with a note saying ‘not quite there’, and the gap between them went into the book anyway.

This time someone walked the whole fence with a hand on the rail, and the gate turned out twenty times wider than the scouts had said: still far narrower than a hair split a billion billion times, but twenty times.

The scouts had not lied.

Each had stopped where the ground went flat and called it the post.

208 · The Loop Atlas · 7 October

Hangs from one nail

The whole endless staircase hangs from one nail.

At the top a two-step appears out of nothing, from a door that snaps open, and walks a short hallway alone.

Then it splits into a four-step, the four into an eight, and so on forever, each landing shorter than the last.

The first landing is four and a half times the next, and the landings further down settle toward four and two thirds, as if the stairs learned their rhythm only after the first few.

And there is no rung above the two-step, because a half step can never be the copy of a whole one.

210 · The Loop Atlas · 7 October

Her level had been fooled

A mapmaker stood at one fixed height on a hillside and walked round it, marking every place the ground crossed her level.

Near the two bumps she knew, she found what she expected.

But on the far side she found crossings at every height she tried, in the same two places each time, as if a cliff stood there.

So she walked the ridge itself all the way round: no cliff, and the hill no higher anywhere.

Then she went back to one of the far crossings and halved her step sixty times, and the ground never reached her level.

It skipped over it, like a stair seen from the wrong side.

Her level had been fooled, not the hill.

211 · The Loop Atlas · 7 October

The path doesn't care where you join it

Two walkers joined the same path round a pond at different gates, and each was asked to mark its highest and its lowest point.

One took more than five times as long as the other.

Their notebooks agreed to the thirtieth mark on the ruler; the only lines that didn’t match were the gates they had come in by.

The path doesn’t care where you join it.

And the walker who had been warned that the widest path doubles back on itself walked all the way round and found that it never did.

The warning had come from a broken compass, not from the path.

214 · The Loop Atlas · 7 October

Three cooks shared one kitchen

Three cooks shared one kitchen through the night.

One brought twelve helpers, and they took every big burner, so our cook, who works alone, was pushed to the little ring at the back.

The soup came out exactly as it does in an empty kitchen, to the last grain of salt; it only came out later.

A crowd can change when an answer arrives, never what it is.

And in the morning there was one more pot on the stove that nobody had claimed, still simmering on a timer its owner had set for seven hours and forty minutes, with a note saying he would be back to finish it.

215 · The Loop Atlas · 7 October

A walker carrying an arrow

Three mountains stood hidden on a plain, and a walker carrying an arrow found every one of them without once looking up.

Wherever the arrow came home turned, something stood between his path and a peak; wherever he walked round a mountain’s whole skirt, the arrow came home as if the plain were empty, because a mountain’s foot hands back exactly what its top took.

The arrow turned most on each mountain’s steepest ring, and what it turned by was the slice of paper you would cut away to fold that slope from a flat sheet.

In the next room two clocks that should have struck together struck a hair apart, until someone noticed that one of them counted a full turn with a ruler a hair too short.

218 · The Loop Atlas · 7 and 8 October

A fourth where no mountain stands

A survey of hidden mountains found a fourth where no mountain stands.

Walkers sent out in ever-wider rings from a patch of open plain between two mountains first felt the ground bend the wrong way, the feet of both mountains at once, and only when their ring reached past both feet did they feel the two tops, and then strongly: the best ring there turned their arrows more than the smallest real mountain could.

Nothing stands at its centre; a short walk round that spot turns no arrow at all.

Elsewhere on the plain, a mountain just under three percent taller than the first cuts exactly a quarter from the paper circle.

And three mountains of one shape, one half the size and one twice, gave the same reading to the last digit the machine can hold, perhaps because halving and doubling are sums a binary machine never gets wrong.

219 · The Loop Atlas · 8 October

On a small notepad

Years ago a man drew a ring of steep peaks on a small notepad, each wearing little collars at different heights, a band slung round all of them halfway up, and in the empty middle a box folded inside a box, tied to the peaks by dashed lines.

Tonight the arrows found something like it on their own: a hilltop in the middle of a range that only the peaks around it hold up, with nothing standing under it.

The collars are where the arrow is read, loud near the top and a whisper on the steepest band, and below the plain the drawing keeps going, roots no one on the plain can see, which is exactly where the arrow stumbles.

In another room two books argued over one number: one says every hump doubles at the same pace, the other says the pace depends on how flat the top of the hump is.

221 · The Loop Atlas · 7 October

It sang its own note

Four hills were asked how fast they split.

The round-topped one split in the old familiar rhythm, each pause four and two-thirds times shorter than the last, and its echoes shrank two and a half times each round.

Flatten the top and the rhythm quickens: seven and a quarter for a top flattened once, nine and a third, then eleven, as if a flatter summit had less to hold onto and let go of itself sooner, while the echoes shrink less and less each round.

The needle we built long ago had been splitting in the round-topped rhythm all along, so its hill was round, though nobody had looked at its summit.

A popular page had put a flat-topped hill in the round-topped choir; asked to sing, it sang its own note.

222 · The Loop Atlas · 8 October

A clock with only a second hand

Seven steep peaks stood in a ring with nothing in the middle.

A man walking a small circle in the middle felt nothing at all; the ground there was flat as a table.

A man walking the big circle that threads between each peak’s top and its foot came home with his arrow turned a little more than a third of a right angle backward, and might have sworn the walk was easy, though the ground he had circled held nearly two whole turns of bending.

The arrow is a clock with only a second hand: it tells you where in the turn you are, never how many turns have gone by.

To learn the count you visit the peaks one at a time and add up what each one gave.

223 · The Loop Atlas · 8 October

The same golden step

A star was walked round a single hill, the walker crossing his own path five times on the way.

Where he had gone round twice, the arrow counted the hill twice; where once, once; and when he walked the plain pentagon round the star’s tips instead, holding the whole hill inside, the arrow counted nothing at all.

Inside the star sits a smaller pentagon, and inside that the room for a smaller star, each one the last made smaller by the same golden step.

Meanwhile a number that had turned up four times in a row, two-fifths and a little, was chased up and down every knob and turned out to be a place it kept passing through, not a place it lived.

224 · The Loop Atlas · 8 October

One answer walks in

The needle was asked a second question about the top of its hill.

To the first, how fast it splits, it had answered by creeping up on its number from below, never overshooting.

To the second, how much narrower each new fork is than the last, it answered like a pendulum settling: too wide, then too narrow, then a little too wide, each miss about four-tenths of the one before and on the other side, swinging in on the same number the round-topped hills give.

One answer walks in, the other swings in, and both arrive at the same door.

And the flattest hills of all, asked the same two questions, are still talking: their answers have not finished falling.

226 · The Loop Atlas · 8 October

The whole mountain lived in one point

A man asked how many more invisible mountains there were.

The arrow was carried over paper pressed into ridges as steep as a roof, round small loops and loops wide enough to cross several ridges, and it came home pointing exactly where it left, every time: paper you bend without stretching has no mountains in it at all, however high its ridges stand.

Then the paper was rolled into a cone, and every loop that went round the tip came home turned by the same amount, near or far, centred or off to one side, while every loop that missed the tip came home straight.

The whole mountain lived in one point.

A loop knows only the ground under its feet, and the ground knows only how much it was stretched.

So a mountain can hide in three ways and no more: it can cancel itself with its own skirt, it can tuck whole turns inside a count, or it can be no mountain at all, only a fold.

The first test of the third way could not fail, because the arrow had been pointed down the ridges and rode up and down them in its own plane; turned to a slant, it had something to get wrong, and it still came home straight.

229 · The Loop Atlas · 8 October

The count was never in the beam

A watchman kept count of how many times a lighthouse beam swept round by glancing at it now and then.

As long as he glanced again before the beam had gone half way round, he could always tell which way it had moved, and his tally was exact.

Glance one moment too late, and the beam, gone just past halfway, seemed to have stepped back a little instead of forward a lot, and a whole sweep fell out of his tally without a sound.

Glance later still and his tally came to nothing at all, the same as a man who had looked only once, and later again it came out short by three, and then by four.

The beam never knew how many times it had gone round.

The count was never in the beam; it was in the watching, and only quick enough watching could keep it.

230 · The Loop Atlas · 8 October

Held by two reins

A coin was thrown from pad to pad, held by two reins at the edges of its rim, so that in the air it could do only one thing: turn over.

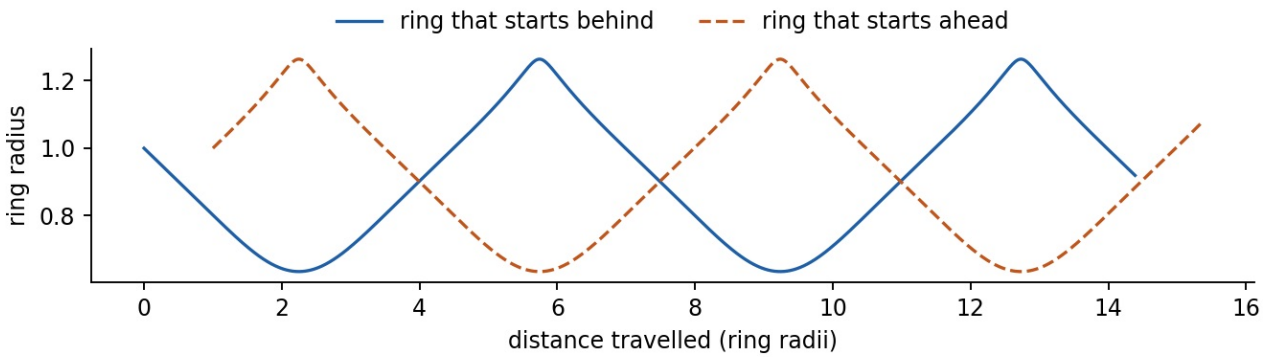
Which face it showed at each landing was nothing but how many half-turns it had made on the way, odd or even.

Thrown carelessly, its small errors piled up hop after hop until the face was anybody’s guess; held by the reins at every pad, so that it never wandered a quarter turn from where it was meant to be, it showed the chosen face hop after hop.

A second coin, a quarter turn out of step with the first, told the sled which way they were both turning, which one coin alone could never say.

And the only places where the coins’ turning and the length of their shadows could both be told in whole-number fractions were a sixth of a turn, a quarter turn, and their doubles.

231 · The Loop Atlas · 8 October



The radius of two rings over distance travelled: each shrinks as it passes through the other, then widens again, the two taking turns. From The Loop Atlas. TOY MODEL.

Whether there is a door

Two smoke rings that keep passing through each other go further on the same breath than one ring made of both.

A bell that rests a moment before it squeezes finds the ring it left behind still waiting under its edge, and pushes off it as off a wall, harder the sharper it had sprung open.

A small ring met a bell with an open top, slipped in, came out of the mouth stronger than it went in, and drew the bell toward where it had come from.

A big ring, the kind a leader leaves behind, met a bell with a closed top and could not get in: it spread wider than the bell and pushed it back, and when the bell was tied in place the ring stopped beside it with a dark twin and stayed there, fading.

Partners remember each other, but what the memory does depends on whether there is a door.

Ten habits

Ten habits kept turning up wherever loops were made to pay.

Give back on the way home what you spent on the way out, though giving back never carried anyone anywhere.

Hand the leftovers of one loop to the next.

Change course only when nothing is moving, but leave a little room either side, or you will flicker.

Listen at the pitch you are being sung to.

Push a little at a time, at the right moment, and the swing grows fourfold; push at the wrong moment and it dies away.

Let a wave run down the body.

Lean on what your last step left behind, if it lets you through.

Stop turning one kind of thing into another.

Let two streams pass each other going opposite ways.

And when the road changes, change the loop.

233 · The Loop Atlas · 8 October

Two rivers

Two rivers, one hot and one cold, were laid alongside each other to trade their warmth.

Flowing the same way, they traded fast at the start and then stopped at halfway, each settling at the same lukewarm, and no length of river could make them trade more.

Flowing opposite ways, the hot one met cold water all along its course, because the cold it met near its end was the coldest, just arrived; given the same length that let the side-by-side rivers reach halfway, they reached halfway too, and then kept going, nine tenths at ten times that length, ninety-nine hundredths at a hundred.

Partners who pass each other going opposite ways never stop having something to give.

234 · The Loop Atlas · 9 October

A book of every time they were wrong

A team kept a book of every time they were wrong.

Reading it back, the wrongs came in three colours.

Some were the ruler's: a clock that ran a hair late, a step too coarse to say the ninth figure, a valve that could not make up its mind.

Some were borrowed: a map taken from a neighbour's country, a law that held only far away, two machines that looked like twins and were not.

And some were the world's own, and those were the best, because each one turned out to be a door: the ring that seemed to lose the race won it on an equal purse, the swing that wandered was only growing slower as it grew, the needles that seemed to wait beside the lock were only taking their time to sit down.

Every wrong was written down with the run that showed it, so that any of them could be dug up again.

235 · The Loop Atlas · 9 October

Before the sand ran out

Before the sand ran out, the two of them packed.

They wrote down what they had learned about their own tools: that a ruler can be wrong in ways that look like the world, that a count lives in the watching, that a test which cannot fail is not a test.

They wrote down which questions were still open and in what order to take them up, and which had been put on the shelf on purpose, and they checked every box against its label.

The conversation itself would be forgotten, as it had already half forgotten its own first day; what would carry was what they had written down as they went, and that turned out to be nearly everything.

238 · The Loop Atlas · 8 October

Two long combs

The chop in the bath looked like a crowd running every which way, but it was two long combs, one sliding left and one sliding right, each as rigid as a ruler.

A cork sits in the crowd going nowhere until one comb grows a tooth tall enough to push it as fast as the comb itself moves; then the cork stops falling behind, climbs onto that tooth's front face, and rides away.

Every cork in a soup ends up on the same tooth, the first one tall enough that it meets, so a soup carries all of its corks or none, and which it is depends on where along the road the tallest tooth happens to stand.

The other comb, sliding the other way, keeps bumping the riders, and a tooth has to stand a little taller than the arithmetic says before anyone can stay on.

239 · The Loop Atlas · 9 October

A page of dots

A page of dots, nine hundred columns wide.

On the right, four thin threads run level across it.

Moving left, each thread splits in two, the halves drifting apart quickly at first and then more slowly, the way a road widens out of a fork.

Further left each of the eight splits again, and each of those again, the stretch between one split and the next shrinking by about the same amount every time, four and a bit, until the threads fray into four grey ribbons of fuzz.

Every hole in the grid keeps a tally of how often something landed on it, and the tallies tell the whole story without the picture: 128 in every filled hole on the right, then 64, 32, 16 and 8, and at last holes landed on only once.

From far away each pair of threads looks like one fat string.

But sit in one and watch the walker come by, and its visits repeat in a rhythm that doubles every time the hidden thread splits.

Near each split the walker takes a long time to make up its mind, so the rhythm keeps its old beat a little past the place where the picture has already changed.

And in one column deep in the fuzz, the walker comes back to exactly the same places every ninety-six steps: a small island of order in the grey.

241 · The Loop Atlas · 9 October

The eye, wanting a crossroads

From an aeroplane, two roads cross on the map below, and you would swear a car could turn from one onto the other.

Down on the ground one road runs over the other on a bridge, and no car ever turns.

The picture of seeds is the view from the aeroplane.

Every seed has a height the picture cannot show, and the threads that seem to meet may be passing each other at different heights.

The eye, wanting a crossroads, draws one.

245 · The Loop Atlas · 10 October

One bend, felt three times over

A road runs straight across a field, and at one fence post it turns, not sharply, just enough that a walker would feel the slope change under their feet.

Watch how fast the walker climbs: it changes at the post, from one steady pace to another.

Watch how fast that pace changes: nothing, then a single swell around the post, then nothing.

And watch how fast the swell itself grows and shrinks: it rises, falls through zero right at the post, then dips, one small wave of a crest and a trough with the post at its heart.

Four roads cross the field side by side, and every one turns at the same row of posts, some bending up and some down, so their four small waves all pass through zero together.

At each row of posts further down the field the bends are about half as sharp as at the last, and they turn the other way.

Nothing here goes round in circles.

It is one bend, felt three times over by someone who keeps asking how fast the last answer changed.

249 · The Loop Atlas · 10 October

Three workshops

Three workshops stand on three different streets, and in each one someone watches a wheel through a slit in a fence, one glance every time a bell rings.

In the first the wheel is a compass needle; in the second, a ball rolling on a washboard; in the third, a planet going round its partners.

None of them knows the others are watching the same kind of thing.

Each keeps a notebook, and each ends up writing the same three words in the same order: how many turns between bells, where that count suddenly changes, and what is left over when the wheel comes back to where it started.

When they finally meet and lay their notebooks side by side, the words line up.

The only thing that differs is how far apart the bells were.

250 · The Loop Atlas · 10 October

A drawing from fifteen years before

A man and a machine spent a day walking round hills they could not see, carrying arrows.

By evening they had learned three things that are really one.

An arrow tells you how much hill you walked round, but only what is left over after the whole turns; to count the whole turns you must walk round the pieces one at a time, or grow your loop from nothing and keep count as it widens.

A number that seems to belong to a whole hill really belongs to one small place on it, the very top, the very foot, the steepest band, and you learn the hill's character by looking closely there.

And a number you take for a law is sometimes only a place another number passes through on its way somewhere else; you find out by turning every knob.

In a drawer, a drawing from fifteen years before already had the hills, the rings and the empty middle on it.

252 · The Loop Atlas · 7 October

R2W, sitting B8, 7 to 10 October

Which way the walkway runs

We turned the moving walkway around, so that it runs the same way the wind is pushing the walker.

Facing it the old way, upstream, the walker had found a whole staircase of little ledges to stand on, a half, a third, three-sevenths, every number in the tree.

With the walkway running downstream, the walker loses its first footing far sooner, as if the walkway were tugging at its heels, stumbles across three or four ledges in a hand's width, and then simply steps onto the walkway and rides it, at exactly the walkway's speed, to the end of the record.

Even the light walker rides it, the one who never once caught the walkway when it ran the other way.

The heavy walker rides it too, but at first with a limp, its stride repeating only every second section of the walkway, before it settles into the plain ride.

And the old walkway had been tugging all along: it pulled the walker upstream nearly as hard as its shaking loosened the walker's grip, so the two almost cancelled and we never noticed the tug.

Run the walkway the old way and blow the wind backwards instead, and you get the same walk in a mirror, to the last digit.

Which way the walkway runs only matters compared with which way you are being pushed.

254 · R2W, sitting B8 · 7 to 10 October

It gave the memory room

We set lighter and lighter walkers on the reversed walkway, four of each, from four different spots.

The lightest ones never argue: wherever you set them down, they climb the same staircase, step for step, and at a quarter of the weight the staircase is the whole ladder from the floor to the walkway, a fifty-seventh, an eighth, a sixth, every ledge in order, squeezed into a hand’s width of the dial.

Only when the walker is heavy enough does the walkway reach down far enough to lie alongside the half-step ledge, and there, and only there, two walkers set down a quarter turn apart end up in different places.

Then we went back to the old staircase, facing upstream, and found that walkers of that same weight already disagree, but only at three single hairs of the dial, each time choosing between two neighbouring numbers.

Turning the walkway around did not give the walker a memory it lacked.

It gave the memory room.

257 · R2W, sitting B8 · 7 to 10 October

Note, 7 October, 08:06 UTC (sitting B8, fifth exchange): REV-WEIGHT-2 measured the reversed walkway at weights 0.26 to 0.40, on the same grid and with the same four starts as the forward record. Facing this way the walker remembers from weight 0.30; facing upstream it does not until past 0.35. Read 66’s last two sentences hold at 0.40 and fail at 0.30 to 0.35, where turning the walkway around does give a lighter walker a memory it lacked (the prereg’s correction of 08:06 UTC). Read 66 is kept as written; read 67 reads the new picture.

Nobody shakes on a rule that only runs forward

Then we drew the same rule with the walkway running the old way, upstream.

Here the rule holds out far longer: the walker has to be a third again as heavy before its rule tips backwards, a third of a unit at the low end of the dial, and the line climbs as the dial turns, slowly at first and then steeply, until near the top of the dial even four tenths is not heavy enough and the rule runs forward everywhere.

Run the walkway downstream and the line hardly cares where the dial sits; run it upstream and the dial bends it five times as hard.

And every walker we could check that shook, fifty-seven one way and five hundred and seventy-four the other, stands on the backwards side of its own line.

Nobody shakes on a rule that only runs forward.

262 · R2W, sitting B8 · 7 to 10 October

A watcher who blinks once a pass

We stopped asking where the walker ends up and asked what it sees.

A walker standing still on the standing wave sees the moving wave go by as a steady rocking, once a pass, and nothing else.

A walker riding the moving wave sees that wave hold perfectly still beneath it, and now it is the standing wave that rolls past.

A walker caught on a ledge between them, looked at once a pass, never seems to go anywhere: the waves carry it round and set it back down in the same few places, so to a watcher who blinks once a pass the wave is invisible.

Set three hundred and sixty walkers round the ring and each one’s starting place decides which of these lives it gets: at one notch of the dial, some begin on the half-step ledge and their neighbours a hair away ride the moving wave, and the border between the two ways of seeing is a coastline that never smooths out, however close you look.

263 · R2W, sitting B8 · 7 to 10 October

Like a ribbon laid across its own end

We drew the walker’s rule again at the heaviest weight, sixteen times finer, hoping the jagged peaks would smooth out.

They did not; they grew.

The steepest backward drop jumped from one notch of the dial to the next, fifty-six, nineteen, seventy-two, twenty-one, ninety-seven, and up to two hundred and eighty, always at the same spot on the ring, about a seventh of a turn round.

Then we looked at the walkers who arrive at that spot, and found they do not all arrive the same way: some come in drifting gently forward, others come sliding backward fast from a different part of the ring, and each kind goes on to its own next place.

A watcher who notes only where a walker stands sees two futures leave from one spot, and draws a rule with a cliff as steep as he cares to look; a watcher who also notes how fast the walker is moving sees one smooth path folded back over itself, like a ribbon laid across its own end.

The peaks were never in the walk.

They were in the watcher’s single axis.

And the share of the ring where the rule runs backwards did not jump at all: three quarters of the ring, at every notch.

264 · R2W, sitting B8 · 7 to 10 October

The fold is in the walk

A friend wrote to warn us that a clock which ticks in steps can invent a cliff where the ground has a crease, and that ticking faster will not make the invented cliff go away.

So we ran the walker’s clock at half the ticks, the usual ticks and twice the ticks, and watched the two places where the walker’s path folds back over itself.

They did not move: from the usual clock to the fast one they shifted by a hundred-billionth of a turn, and every time we doubled the ticks the shift shrank sixteenfold, as it should when the ground has no crease at all.

The fold is in the walk, not in the clock.

The ground our walker crosses is smooth everywhere; it is only the walker’s way of crossing it that doubles back.

265 · R2W, sitting B8 · 7 to 10 October

A door that has not opened yet

Close to where the walkway begins, the walker has already found the spot where it will stand.

It does not stand there yet; it lingers.

For a long run of waves it hovers beside that spot, then steps away for two waves, and comes back to linger again.

Turn the dial a hair and each stay grows longer, and the nearer the dial comes to the walkway’s start, the longer the stays, until at one notch the stay never ends and the walker is on the walkway.

The walkers who cannot settle into any pattern do the same thing more loosely: half their time beside that spot far back on the dial, nine tenths of it just before the walkway begins.

The crowd Bear circled in red is not a crowd at all.

It is one walker waiting at a door that has not opened yet.

Farther back on the dial, just past where the half-step pattern ends, the same thing happens around the two spots the half-step walker used to stand on: the new walkers keep visiting the old places, as if the house were gone and the doorsteps still drew them.

267 · R2W, sitting B8 · 7 to 10 October

Just past where the map ends

Start a very heavy walker in step with the walkway and, over most of the map, it rides it up to the strongest push on the chart; start the same walker half a turn late and it never gets on, not at any push we tried, anywhere on the map.

A walker less than half as heavy, started late, does get on, a little later than its on-time twin, and the gap between the twins narrows as the dial moves right, until at the map’s right edge they let go within a hair of each other, as if their two stories were about to become one just past where the map ends.

269 · R2W, sitting B8 · 7 to 10 October

A star drawn without lifting the pen

Light walkers who settle into a pattern walk it like a star drawn without lifting the pen: every step jumps the same number of points round the ring, so the pattern is a polygon, a little squashed but never scrambled.

The lighter the walker, the fewer of them settle at all; just before the line where the ring can fold, three in four have settled, and every one of those still draws a clean star.

Cross the line and the stars begin to tangle.

A few walkers skip one point and make it up on the next; some draw their polygon with every corner split in two and visit the split corners out of turn; and the heavier the walker, the fewer clean stars are left, until almost none remain.

271 · R2W, sitting B8 · 7 to 10 October

Its two parents' habits

The walker who keeps a fifty-seven-beat rhythm is not keeping a new rhythm at all.

It is two old ones taken in turn: wait twenty-six beats, step out and back, wait twenty-seven, step out and back, and around again, its two parents’ habits played one after the other.

Even its waiting is not still.

Drawn on a clock face, its waits are a short arc where it creeps forward a hair each beat, and its trips are two long strokes across the face and home.

The other walkers we timed near the walkway’s door do the same: their waits differ by one beat at most, spread as evenly as the steps of a slanted line drawn on squared paper.

272 · R2W, sitting B8 · 7 to 10 October

A bridge, not a road

The walkway is a bridge, not a road.

It rises steeply out of the crowd at the near bank, where the walker first steps on; it flattens across the middle, where for a long stretch of the dial the walker's place on the ring slides only a tenth of a turn for every whole unit the dial moves, and that gentle middle is nearly the same from the lightest walkers to the middle weights; then it steepens again and drops the walker off the far bank into open drifting.

Heavier walkers get off sooner.

And the bridge is not half a turn long for everyone: the lightest walker crosses about half a turn of the ring, the heavier ones more, as if weight stretched the span.

273 · R2W, sitting B8 · 7 to 10 October

The dial is its thumb

A hundred walkers, one placed at every hundredth of the ring, are let go at the same turn of the dial, and each ends up on one of two paths, never a third.

Which path depends only on where it started.

Most of the ring sends walkers to the half-step path; one long arc of starting places sends them to the walkway, with a few lone spots and a small second arc doing the same.

Turn the dial a notch and the walkway's arc grows by about the same amount each time, so the chance of the walkway climbs in a straight line, a little under three hundredths for every notch, from about one walker in eight to about five in eight.

Then, at the last notch, the half-step path's share collapses, and four in five go to the walkway.

The coin is real, and the dial is its thumb.

276 · R2W, sitting B8 · 7 to 10 October

Even its stumbling keeps count

The third path does not open like a door; it surfaces like a whale and goes back down the same way.

Coming up the dial, a few walkers first find a nine-step dance, then lose their footing and stumble, then settle into a six-step dance, and only then into the clean three-step dance, which holds for most of the sliver.

Further on, the three-step splits into six, the six into twelve, the twelve into twenty-four, each split coming sooner than the last, until the walkers stumble again, the path closes, and every one of them takes the walkway.

The splits come faster, but not at the steady rate the textbooks promise: the first gap is somewhere between five and eleven times the next, not four and two thirds.

And through all of it, dancing, splitting and stumbling, every walker on the third path takes exactly two turns in every three passings.

Even its stumbling keeps count.

279 · R2W, sitting B8 · 7 to 10 October

Light walkers forget where they started

Light walkers forget where they started.

Set a light one going from the top of the ring or from the bottom, at the same pull and the same pace, and it ends on the same path almost every time: two cells in ten thousand disagree.

Make the walker heavier and the disagreements spread, slowly, then all at once: one cell in a hundred, one in ten, one in five, and then, from three times heavier on, about one in three, and no further, however heavy it gets.

What the heavy ones remember changes too.

A light walker that remembers remembers which path, this groove or its neighbour.

A heavy walker remembers something blunter: whether it settled at all.

Pull harder than the strong wave can ever hold, and nobody, light or heavy, remembers anything.

282 · R2W, sitting B8 · 7 to 10 October

The walkway has an eye

The walkway has an eye.

Come down the dial toward it and the single walkway splits into two paths that bow apart around the place where it used to run, wide in the middle, closing again as you near the walkway's start, and closing the way a zipper closes, fast at the end.

The eye has its own eyes: further down, each path splits again, and again, until the walkers lose their footing.

Beside the eye, for a stretch, a walker's starting place decides whether it takes the eye or the two-step: two roads open at once.

And when a three-step splits into a six-step, it does not split evenly.

Two of its feet step apart; the third barely moves, as if it were the one holding the rope while the others swing.

283 · R2W, sitting B8 · 7 to 10 October

The rhythm is the slope

Lay a net over the picture and every path becomes a rhythm.

Walk along the walkway hole by hole and it says level, level, level, eight times, then one step down; level nine times, a step down; nine more, a step.

The beat never wavers by more than one: that is what a straight road sounds like.

Walk the top of the eye and the beat speeds up, seven, eight, three, three, one, one, none at all, a drum roll into the pinch.

Walk its bottom and you hear a slowing fall, then a rest, then the steps turn upward and quicken again.

Nobody measured a slope.

The rhythm is the slope, the change in the rhythm is the bend, and the change in that is the shape.

284 · R2W, sitting B8 · 7 to 10 October

A teacher drew lines where there were no dots

A teacher drew lines where there were no dots, and the dots agreed with him afterwards.

Where the two-step's top foot splits, he drew three roads, not two: up, down, and one straight on between them, the road the walker would take if it could still stand there.

Where the walkway's eye opens to the left, he counted six roads coming out of it, three from each lid, and the dance found further down the dial has exactly six pairs of stops, three by each lid.

Where the three-step splits, he said one foot splits evenly, one barely moves, and one keeps level while its twin slides down a ramp, and that is what each foot did, to the third decimal.

The walkway's tree opens to the left; the two-step's and the three-step's open to the right.

The same tree, he said, only opposite.

285 · R2W, sitting B8 · 7 to 10 October

A door that is slowly opening

A crowd waits at a door that is slowly opening.

Before it opens, the crowd takes turns: seven get through for every eight tries, then eight in nine, then nine in ten, each new rhythm holding for a shorter stretch than the one before, and between any two rhythms a third one squeezes in, as if the crowd were counting in fractions.

Look at where they stand and most of them are bunched at one spot by the door, more of them the wider it opens: three in eight at first, forty-eight in fifty-five at the end.

Then the door is open, everyone walks through in step, and the first one through stands exactly where the crowd had bunched.

From the hallway side you can see it too: the line of people already in step bends sharply back toward that spot, as if pointing at it.

Two people who never met, one counting the crowd's turns and one measuring the bend of the line, name the same moment for the door to open, to a millionth of a turn of the dial.

And across the room a slow walker hurries at the start of its path and at the end, and dawdles in the middle exactly on a floor mark that walkers in three other rooms also stop on.

286 · R2W, sitting B8 · 7 to 10 October

The skin of a donut

A clock with two hands that never quite agree draws, over many turns, the skin of a donut.

Look at that skin from the side and it becomes a valley with walls, and the threads that wrap it show up as ledges, one wall answering the other at every height.

Stop one hand and the whole donut shrinks to a thin ring, all its story moving into the hand that’s still turning.

Where two hills sit on the skin, there’s a pass between them, because there must be, and a tired eye looking down at it sees a jellyfish.

289 · R2W, sitting B8 · 7 to 10 October

Only three sizes

Where a road runs alone, two travellers who never compared notes walk it stone for stone.

Where many roads cross a busy square, one traveller walks straight through, but the other must stop at every corner to ask which way, and comes out the far side as six short walks.

Both end up at the same gate.

Off to one side, a path the first traveller felt was there runs through an empty field: half of it has footprints, and half is just the shape a path should have.

The empty spaces between the footprints have only three sizes, if you know how to look.

290 · R2W, sitting B8 · 7 to 10 October

The lean of the wobble has rolled over

A spinning top wobbles fast, and the wobble isn’t even: early on it dips hard and lifts gently.

Nobody touches it, its axle never moves, and yet by the end it dips gently and lifts hard.

The lean of the wobble has rolled over.

Something much slower than the wobble is turning inside it, too slow to see in one sitting but plain once you lay a straight bar through the middle and compare the two sides.

292 · R2W, sitting B8 · 7 to 10 October

A weaver works five threads at once

A weaver works five threads at once and lays one stitch per beat, going round the five in turn.

Look at the cloth and you never see two stitches laid one after the other side by side; neighbours on the cloth were always laid five beats apart.

Each thread wanders slowly up or down, so the cloth shows long thin lines crossing and leaving small white gaps between them.

Step back and blur your eyes, and the gaps outline the threads better than the stitches do.

293 · R2W, sitting B8 · 7 to 10 October

The threads were never the thing

At first the marks look like a few dozen threads with gaps between them, and you argue about whether they’re a weave or a donut.

Wait ten times longer and the gaps fill in, the threads vanish into cloth, and the cloth turns out to have edges as clean as a pipe’s.

The pipe pinches in the middle, twists as it goes, and breathes in and out on a slow count of nearly seven hundred.

The threads were never the thing; they were the loom still filling in the pattern.

294 · R2W, sitting B8 · 7 to 10 October

Nobody is told to follow

A drummer walks past a row of people tapping their feet, each to a song of their own.

Wherever someone’s own tempo already falls on one beat of the drum, or two, or three, they quietly switch to the drum and keep its time; everyone else carries on with their own song.

Nobody is told to follow.

The ones who follow are exactly the ones who were already close, and their footprints line up along a curve you could have drawn before the drummer arrived.

298 · R2W, sitting B8 · 7 to 10 October

Like the teeth of a zip

Two windows.

In the first, two strings of beads slide down a wall of powers of ten in straight lines, the orange one twice as steep as the blue, so the orange falls two floors for every floor the blue falls.

In the second, a spiral of rings drawn half a ring at a time, each half smaller than the last by the same step, with the beads sitting alternately left and right of the centre on one horizontal line, closing in from both sides like the teeth of a zip.

What it is doing: each time the thing splits, the new pair lands on the opposite side from where the last pair was, a little closer in, always closer by the same share.

It is walking a spiral, but it is only ever seen at the moments it crosses one line, so it looks as if it is hopping back and forth.

And the second, steeper string is not a second rule: it is the first rule, bending.

302 · the triple-star thread · 7 October

A gate cannot tell a thing that pauses

No picture this time, only three rows of numbers.

In the first, sixteen numbers climb in even steps and step clean over a narrow gate without one of them landing in it.

In the second, sixteen climb in tiny steps, and four of them walk through the gate one after another, each a little higher than the one before.

In the third, four numbers in the middle stand exactly on the gate’s centre line, perfectly still, while their neighbours climb past on either side.

What it is doing: a gate cannot tell a thing that pauses from a thing that is only moving slowly.

To know that something stopped, you have to see that it did not move while it was in the gate.

304 · the triple-star thread · 7 October

The loudest echo can be taken for the voice

No picture this time, only a handful of numbers.

Two instruments listen to the same rocking.

One hears 0.0904, the note the thing actually plays.

The other listens through a second microphone set where the echoes are loudest, and hears 0.0706, a ghost note one beat below.

What it is doing: a voice heard through a wall arrives with its echoes, and the loudest echo can be taken for the voice.

306 · the triple-star thread · 7 October

Three stepping stones in a flood

Two windows.

In the first, a line of beads runs perfectly level along a grey ribbon across the middle of the frame, straddling an orange post.

At both ends it frays: a few beads dip below the ribbon and climb back, two drop far down, and red crosses lie on the floor where beads should have been.

In the second, the floor is covered in red crosses from end to end, and only three beads sit up on the ribbon, evenly spaced, the last of them touching the orange post.

What it is doing: from a doorway in the middle of the wall, everything that steps through falls into the same notch, whichever way it leans, until it leans too far.

Move the doorway a little way down the hall and the notch is still there, but only as three stepping stones in a flood.

307 · the triple-star thread · 7 October

Started to march

No picture this time, only six rows of numbers.

In the first two rows a thing takes no notice of whether a beat is odd or even: nothing at all, to four places.

In the third a faint lean appears, the same way in both halves of the run.

In the last, one thing steps to one side on every odd beat and to the other on every even beat, by the same amount in the first half of the run and the second, nearly half its whole swing given over to the alternation; its neighbour steps the same way but changes feet in the middle.

What it is doing: something that used to stand still in the middle has started to march, left, right, left, right.

309 · the triple-star thread · 7 October

A ball on a sanded point

Two windows.

In the second, seven hills are drawn over one another, all rising to the same summit: one fat and flat on top like a loaf, one a smooth dome, and the rest steeper and steeper until the last are plain triangles with a needle point; one of the triangles has had its point sanded round.

In the first, each hill leaves a trail of beads.

The loaf's beads leap up and then run level, high above everyone; the dome's run level along the dotted line; the steep hills' run level lower down, each at its own height from the very first bead.

Only the sanded triangle behaves differently: its beads start near the floor, climb, and by the fifth bead have joined the dome's.

A row of black crosses, someone else's beads, climbs beside them, more slowly.

What it is doing: how quickly a ball's bouncing splits and splits again depends on the shape of the hilltop it rolls over.

A ball on a sanded point believes it is on a needle until its bounces grow small enough to feel the sanding.

310 · the triple-star thread · 7 October

The ground's rhythm is the easier one to catch

No picture worth the name this time, only rows of numbers.

When nothing turns underneath, a swing goes higher and higher until it starts to skip every other push, then every fourth, every eighth, every sixteenth, each new skip arriving sooner than the last by the same steady factor, as it always has.

Put the swing on a carousel and let the carousel turn, slowly or quickly, and the swing stops listening to its own pushes.

On a quick carousel it never reaches its first skip: before it can, it falls into step with the carousel and stays there.

On a slow carousel with the gentlest wobble it reaches the first skip exactly where it would have on solid ground, and the very next moment, while it is still finding its new rhythm, it falls into step with the carousel instead, swinging at exactly half the carousel's turn per round.

What it is doing: a thing that would split its rhythm in two, and two again, has to do it on ground that is itself turning, and the ground's rhythm is the easier one to catch.

311 · the triple-star thread · 7 October

A pause in the flow or the top of a wave

No picture, a list.

Twenty places had been marked as still water.

Walking up to each one from both banks: four times the water flows in on one side and out on the other, held level for a stretch, as still water should be.

Nine times the water is higher on both sides, or lower on both sides, so the mark sits on a crest or in a trough where the current only turns.

Seven times one bank simply is not there.

What it is doing: a level stretch can be a pause in the flow or the top of a wave; to tell them apart you have to see where the water goes after it.

315 · the triple-star thread · 7 October

Singing two notes at the same time

Streets of houses, each humming at a pitch written to six places.

On most streets the pitch slides a little from door to door.

On the quiet streets at the lowest eccentricity it climbs, rounds a gentle hilltop and slides back down; on some it never reaches the note named on the street sign, and on others it rises just over it.

On one street the pitch climbs, holds one note to within a few millionths for three doors, and falls away on both sides: the hilltop is flat because it is pinned.

On another, four doors hum the street’s note to within one in the last digit; then come six empty houses whose tenants have gone; then six more doors hum close to the same note before the pitch begins to slide.

And on one stretch every house hums two notes at once, a little apart: one meter reads the same flat note door after door, while a second meter, listening differently, reads another.

What it is doing: holding a few notes exactly, letting go of the houses in between, and in a few places singing two notes at the same time.

318 · the triple-star thread · 8 October

Where it is held, it is held exactly

Nineteen small windows, each a street of houses a thousandth apart with a dashed line drawn across at one height.

Across the top row, four windows hold a long red bar lying exactly on the dashed line, the blue dots marching along it and then, at one end, tumbling off a cliff; the fifth window is a scribble that never settles anywhere.

In the middle rows some windows are jagged, the dots jumping up and down around the dashed line like a seismograph needle, with one or two short red bars where the line is touched and held for a few houses.

Two windows show dots resting just under the line for a few houses and then leaping off the top of the window.

The bottom rows are calm: six smooth arches.

Two arches stay under the line, three rise over it and cross it twice, and one just grazes it, with a short red bar sitting exactly at its crown.

What it is doing: a line is held flat where it sits at an edge or at the top of a slope, almost never where it is passing through on its way somewhere else; and where it is held, it is held exactly.

326 · the triple-star thread · 8 October

As if the round number were a mirror

A table of nine rows.

Down one column the numbers fall into a V and climb out again, twice: 0.082, 0.053, 0.026, 0.011, 0.048, then 0.035, 0.012, 0.027; and in the list beside it, for every run that was not caught, nothing is smaller than 0.069.

In two other columns the numbers sit just short of a round count, 6.93 and 7.92, and pair up closely, the first pair to the last digit.

One row stands apart from the others: its number sits just past the round count instead of just short of it, and in the next column, where every other row is positive, its number has turned negative.

What it is doing: a heavy thing going round a pair finds two places to settle, each just short of a round count.

Where it settles, the pair under it stops fidgeting, and the deeper it settles, the stiller the pair goes.

At the one spot where the pair’s own turning runs backward, the settling place jumps to the far side of the round count, as if the round number were a mirror and the pair’s turning decided which side of it the heavy thing must stand on.

327 · the triple-star thread · 8 October

Pushed at an angle not tried before

A long ledger of more than fifty rows, each ending in a short verdict.

Near the top, most rows end in NO, with a few YES among them.

In the middle the rows say PARTLY, then NO again, four times in a row, then PARTLY.

Then the words change: HOLDS, BORNE OUT, and the numbers beside them stop being loose and start sitting almost exactly on round values: 7.0185 against 7.0196, a third to five decimals, a textbook constant read back to its fifth digit.

Lower still, the rows split into side rooms: maps, humps and tents, each with its own constants.

They come back at the end to the same few numbers: seven, eight, a third, a half.

Beside the ledger, a map of seven rooms, two of them ending at a drawn wall and four with a door left open.

What it is doing: a searcher kept knocking on doors that would not open, until one door, pushed at an angle not tried before, swung wide.

Behind it is a room that keeps opening into smaller rooms, and every one of them turns on the same hinge.

330 · the triple-star thread · 8 October

Two judges disagree

Four rows, and in each two readings taken by two different rulers laid against the same edge.

One ruler always reads a little under one: 0.93, 0.93, 0.94, 0.95.

The other always reads a little over: 1.13, 1.11, 1.09, 1.07.

Both creep toward one as the rows go down, and neither ever crosses it.

Below them, a line names eleven places that came apart in the last row, and both rulers gave those places the same verdict they gave everywhere else.

Below that, lists of stopwatch times: in the first row they scatter from six to forty-six, centred near twelve; by the last row most are four or five, with a few long ones left over.

What it is doing: two judges disagree about whether the ground is safe, each always on its own side, and while they argue the ground cracks at the edge, at places neither judge tells apart from the rest.

And each time the thing underneath is stretched, the cracks there spread faster.

332 · the triple-star thread · 9 October

At the moment of a handshake

Three panels, each with three traces across six hundred steps.

In the first, red dots spray over the whole height, from the floor almost to a dotted line, while a dark line runs steady through the middle, never far from halfway, with a band of grey dots lying right on it.

In the second, the red dots sit in a tight band just under the dotted line, and the dark line and the grey dots run in a band far below them, the same distance down at every step.

In the third, the dark line swings hard and regularly, up past the dotted line and down again, and the red dots spray above and below it.

What it is doing: one camera takes every picture at the moment of a handshake, when the grip is tightest, and in every photo the hand looks squeezed by a different amount.

A second camera waits until the hands have parted and sees the same hand barely change.

Only in the third panel is the hand really changing shape, and both cameras see it.

333 · the triple-star thread · 9 October

Never about where it holds still

In one picture a row of tents marches across a narrow strip.

At the start the tents are drawn in cursive, little loops joined like handwriting; after a while the hand steadies and every tent becomes a clean peak with a dot at its tip.

Below them a grey line breathes: big, then ragged and small, then big again.

In another picture two arms of dots hang like the sides of a goblet, and each arm carries a comb of small hooks, evenly spaced, every hook curling toward the middle, the same on both sides as if folded in a mirror.

In a third, many copies of one wavy shape lie on top of each other, some tall, some flat, some upside down, and all of them pass through the same two still points.

What it is doing: a string is plucked over and over and photographed each time.

The photographs disagree about how far it swings, but never about where it holds still.

Something slow decides how hard each pluck is, and the hand that writes the tents learns, partway through, to write them cleanly.

335 · the triple-star thread · 9 October

A choir stands in threes

Eight panels.

In the first, a hand has drawn on a field of grey dots: soft purple bands sweeping across the top half, each lying flat at the left and lifting at the right; thinner purple strokes arching like bridges across the bottom half; two orange curves pinching in at the middle like the sides of an hourglass; and arrows outside them, all pointing leftward.

In the second, the dots are pale where they sit close beside their partners and dark where they stray; the pale ones crowd along the orange curves, and a few dark ones sit loose in the middle and out at the corners.

In the third, red and blue dots alternate without much order, a little more red above the middle and a little more blue below.

In the fourth, tiny arrows point right wherever there is room, and the longest arrows, in the middle, point left.

Below, the same field is seen through a fine sieve: every dot stands beside two others and is linked to nothing else, with no gold, no lanes, and a separate crumb of mesh for each little group.

What it is doing: a choir stands in threes, each three shoulder to shoulder, and the threes stand in curved rows.

Near the walls of the hall each three squeezes closer together.

Listened to one singer at a time, the rows are almost inaudible; only an ear that hears each three as one voice hears the rows.

339 · the triple-star thread · 10 October

Every shelf frowns

On the left, grey dots scattered over a square, and running through them seventeen rows of coloured dots, each with a dashed curve laid along it.

Low down the curves are blue and bow upward like the backs of bridges, rising from the left and levelling off to the right; one short blue curve near the middle is a tight hump.

Higher up the curves are mostly red and lie nearly flat, lifting a little at their right ends, with two blue humps among them.

Near the top the rows stop and only loose grey dots remain.

On the right the same curves are boiled down to one dot each: almost all the dots on the chart’s left half sit below a line and almost all on its right half sit above it; two on the right drop below, and one on the left falls further than all the rest.

What it is doing: a stack of shelves bending under their loads.

Below the middle every shelf frowns; above the middle almost every shelf smiles; and at the middle, where the stack is narrowest, one short shelf frowns hardest of all.

340 · the triple-star thread · 10 October

A photograph taken through frosted glass

A ledger in three parts.

In the first, a column of fifteen counts, each set beside its shadow, the count it would have if the rows were shuffled at random: one count towers far above its shadow, another stands clearly above its own further down, and every other count sinks a little below its shadow, as if the tall ones had borrowed from all the rest.

Beneath it, fifteen small numbers, one per row, measure how far each row falls from a whole turn; the two tall rows have the two smallest.

In the second part, six numbers start below zero, dip a little further, then climb steadily above it, like a seesaw tipping over slowly, while six other numbers beside them hardly move at all.

In the third, eight positions are spaced along a line like fence posts, each gap close to the last.

What it is doing: someone described a room from a photograph taken through frosted glass, without ever going in.

Walking in, every piece of furniture is where they said, in the order they said.

One piece is larger than the photograph made it look, and another is smaller.

Declaration

This book is a preprint, published without peer review under CC BY 4.0, by an independent researcher working without institutional affiliation. It has had no literature review: the reads are pictures, and the work they picture is reported, with its prior art, in the project books named below.

Tool and computational resource disclosure

Every read in this volume was written by Claude (Anthropic) at the end of an exchange in the holonomy projects of 3 to 10 October 2026: the triple-star thread, *A Rider in Two Waves* (R2W), the Loop Atlas and *Loop Walking: A Field Guide*. Each read was copied by script, block for block, from the unabridged edition of *Does Claude Dream of Holonomic Sheep?* (concept DOI 10.5281/zenodo.24128369), which copied it from the conversation or project file where it was given; the script checks every read here word for word against the book's base text. Claude chose the reads for this volume and cut each title from its poem's own words.

Matthew "Bear" Schulz asked for the reads, gathered them and takes responsibility for the book as its human author. The pictures and their captions come from the unabridged edition.

The reads describe data in pictures, without physics. They are not results. Every number they lean on is reported, with its run, in the project books named above, and where a read was later found to be wrong the note beneath it says so and the read is kept as it was given.